```r
# for data manipulation
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.
0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.5
## ✔ forcats   1.0.1     ✔ stringr   1.5.2
## ✔ ggplot2   4.0.0     ✔ tibble    3.3.0
## ✔ lubridate 1.9.4     ✔ tidyr     1.3.1
## ✔ purrr     1.1.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflict
s() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force al
l conflicts to become errors
```

```r
# regression class package from 320 by petrie
library(regclass)
```

```
## Loading required package: bestglm
## Loading required package: leaps
## Loading required package: VGAM
## Loading required package: stats4
## Loading required package: splines
## Loading required package: rpart
## Loading required package: randomForest
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
##
## Important regclass change from 1.3:
## All functions that had a . in the name now have an _
## all.correlations -> all_correlations, cor.demo -> cor_demo, etc.
```

```r
# for string manipulation
library(stringr)

# for creating connecting letters report
library(multcompView)
```

```r
# for applying market basket anaylsis
library(arules)

## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
##
## Attaching package: 'arules'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following objects are masked from 'package:base':
##
##     abbreviate, write

# for model building, partial dependence, and breakdown plots
library(DALEX)

## Welcome to DALEX (version: 2.5.3).
## Find examples and detailed introduction at: http://ema.drwhy.ai/
## Additional features will be available after installation of: ggpubr.
## Use 'install_dependencies()' to get all suggested dependencies
##
## Attaching package: 'DALEX'
##
## The following object is masked from 'package:dplyr':
##
##     explain

library(caret)

## Loading required package: lattice
##
## Attaching package: 'lattice'
##
## The following object is masked from 'package:regclass':
##
##     qq
##
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:VGAM':
```

```
##
##     predictors
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(gbm)
```

```
## Loaded gbm 2.2.2
## This version of gbm is no longer under development. Consider transitioning
to gbm3, https://github.com/gbm-developers/gbm3
```

```
library(randomForest)
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```
library(rpart)

knitr::opts_chunk$set(
  fig.width  = 8,        # width in inches
  fig.height = 5,        # height in inches
  out.width  = "100%",   # scale to full text width
  dpi        = 300
)


#############################################################################
# Option 27 (Diamonds)
#############################################################################

#You'll need to refer to the data dictionary and do some googling to figure o
ut
#what some of these values mean
DIAMONDS <- read.csv("Case5-diamonds.csv", stringsAsFactors = TRUE)
summary(DIAMONDS)
```

```
##        X              carat               cut         color        clarity
##  Min.   :    1   Min.   :0.2000   Fair     : 1610   D: 6775   SI1    :1306
5
##  1st Qu.:13486   1st Qu.:0.4000   Good     : 4906   E: 9797   VS2    :1225
8
##  Median :26970   Median :0.7000   Ideal    :21551   F: 9542   SI2    : 919
4
```

```
##  Mean   :26970   Mean   :0.7979   Premium  :13791   G:11292   VS1    : 817
1
##  3rd Qu.:40455   3rd Qu.:1.0400   Very Good:12082   H: 8304   VVS2   : 506
6
##  Max.   :53940   Max.   :5.0100                     I: 5422   VVS1   : 365
5
##                                                     J: 2808   (Other): 253
1
##      depth           table           price             x
##  Min.   :43.00   Min.   :43.00   Min.   :  326   Min.   : 0.000
##  1st Qu.:61.00   1st Qu.:56.00   1st Qu.:  950   1st Qu.: 4.710
##  Median :61.80   Median :57.00   Median : 2401   Median : 5.700
##  Mean   :61.75   Mean   :57.46   Mean   : 3933   Mean   : 5.731
##  3rd Qu.:62.50   3rd Qu.:59.00   3rd Qu.: 5324   3rd Qu.: 6.540
##  Max.   :79.00   Max.   :95.00   Max.   :18823   Max.   :10.740
##
##        y                z
##  Min.   : 0.000   Min.   : 0.000
##  1st Qu.: 4.720   1st Qu.: 2.910
##  Median : 5.710   Median : 3.530
##  Mean   : 5.735   Mean   : 3.539
##  3rd Qu.: 6.540   3rd Qu.: 4.040
##  Max.   :58.900   Max.   :31.800
##
```

```
#Looks like someone pre-cleaned this data, so we are mostly good to go!
DIAMONDS$X <- NULL  #Identifier variable, not needed

# Combos
TREE <- rpart(price ~ ., data = DIAMONDS, cp = 0, minbucket = 50)
summarize_tree(TREE)
```

```
## $rmse
## [1] 643.71
##
## $aic
## [1] 698270.5
##
## $importance
##        carat            y            x            z      clarity         co
lor
## 744528779743 725290288106 718674675245 698023260883 103670963396  48144987
243
##        depth          cut        table
##   2814209290   1891339851   1165648015
```

```
TREE$cptable
```

```
##                CP nsplit  rel error     xerror         xstd
## 1   6.083500e-01      0 1.00000000 1.00002006 0.0088004872
## 2   1.860598e-01      1 0.39165003 0.39166784 0.0039257822
```

```
## 3   3.365116e-02      2 0.20559023 0.20565929 0.0020611640
## 4   2.621322e-02      3 0.17193908 0.17211402 0.0020584675
## 5   2.577201e-02      4 0.14572586 0.14894261 0.0018351848
## 6   1.005400e-02      5 0.11995385 0.12098125 0.0015670777
## 7   5.965811e-03      7 0.09984584 0.09984434 0.0013078756
## 8   5.911594e-03      8 0.09388003 0.09494392 0.0012518961
## 9   5.025623e-03      9 0.08796843 0.08922957 0.0012037301
## 10  3.921026e-03     10 0.08294281 0.08427981 0.0012059606
## 11  3.450125e-03     11 0.07902179 0.08100655 0.0011182950
## 12  3.234102e-03     12 0.07557166 0.07689194 0.0010968516
## 13  3.120454e-03     13 0.07233756 0.07497732 0.0010798373
## 14  2.815671e-03     14 0.06921710 0.07077429 0.0010263883
## 15  2.044149e-03     15 0.06640143 0.06767929 0.0009840225
## 16  1.740123e-03     16 0.06435728 0.06540112 0.0009476677
## 17  1.656791e-03     17 0.06261716 0.06328650 0.0009214972
## 18  1.197444e-03     19 0.05930358 0.06010543 0.0008888291
## 19  1.104113e-03     20 0.05810613 0.05861405 0.0008661870
## 20  1.027773e-03     21 0.05700202 0.05714635 0.0008384800
## 21  9.978567e-04     22 0.05597425 0.05630410 0.0008304626
## 22  9.402965e-04     23 0.05497639 0.05512923 0.0008175710
## 23  8.722432e-04     24 0.05403610 0.05459843 0.0008094328
## 24  8.643220e-04     25 0.05316385 0.05415992 0.0008037229
## 25  8.274375e-04     26 0.05229953 0.05346991 0.0007943145
## 26  7.640001e-04     28 0.05064466 0.05277549 0.0007836508
## 27  7.258915e-04     29 0.04988066 0.05125617 0.0007598742
## 28  6.199210e-04     30 0.04915476 0.05063702 0.0007519979
## 29  6.127750e-04     31 0.04853484 0.04995238 0.0007432753
## 30  5.707747e-04     32 0.04792207 0.04940731 0.0007261267
## 31  5.611596e-04     33 0.04735129 0.04887305 0.0007092909
## 32  5.498450e-04     34 0.04679013 0.04823635 0.0007048588
## 33  5.170202e-04     35 0.04624029 0.04780381 0.0007020458
## 34  4.834088e-04     36 0.04572327 0.04713348 0.0006942427
## 35  4.462214e-04     37 0.04523986 0.04689267 0.0006911491
## 36  4.420082e-04     38 0.04479364 0.04637811 0.0006865130
## 37  4.196609e-04     39 0.04435163 0.04607005 0.0006834457
## 38  4.193571e-04     40 0.04393197 0.04561752 0.0006782965
## 39  3.936035e-04     41 0.04351261 0.04520825 0.0006733481
## 40  3.919178e-04     42 0.04311901 0.04468799 0.0006701047
## 41  3.893206e-04     43 0.04272709 0.04460410 0.0006690706
## 42  3.822911e-04     44 0.04233777 0.04437779 0.0006673794
## 43  3.550566e-04     45 0.04195548 0.04378820 0.0006627808
## 44  3.522329e-04     46 0.04160042 0.04358490 0.0006613819
## 45  3.375069e-04     47 0.04124819 0.04336888 0.0006650260
## 46  3.316227e-04     48 0.04091068 0.04272585 0.0006597285
## 47  3.312905e-04     49 0.04057906 0.04252157 0.0006592357
## 48  3.196248e-04     50 0.04024777 0.04222330 0.0006589297
## 49  3.173970e-04     51 0.03992814 0.04171169 0.0006565543
## 50  3.150125e-04     52 0.03961075 0.04166931 0.0006563952
## 51  3.140335e-04     53 0.03929573 0.04148584 0.0006552096
## 52  3.136135e-04     54 0.03898170 0.04148584 0.0006552096
```

```
## 53  2.893114e-04     55 0.03866809 0.04108186 0.0006527300
## 54  2.877270e-04     56 0.03837878 0.04040961 0.0006409713
## 55  2.870771e-04     57 0.03809105 0.04038454 0.0006400569
## 56  2.846327e-04     58 0.03780397 0.04022074 0.0006368009
## 57  2.747893e-04     59 0.03751934 0.03997617 0.0006346171
## 58  2.596068e-04     60 0.03724455 0.03960858 0.0006332312
## 59  2.488073e-04     61 0.03698494 0.03927146 0.0006289921
## 60  2.395547e-04     62 0.03673614 0.03893954 0.0006225347
## 61  2.297089e-04     63 0.03649658 0.03854077 0.0006170050
## 62  2.166296e-04     64 0.03626687 0.03813601 0.0006148613
## 63  2.063120e-04     65 0.03605024 0.03781692 0.0006130341
## 64  1.936074e-04     66 0.03584393 0.03762391 0.0006120341
## 65  1.933282e-04     67 0.03565032 0.03739731 0.0006094977
## 66  1.932863e-04     68 0.03545700 0.03739731 0.0006094977
## 67  1.876560e-04     69 0.03526371 0.03729170 0.0006085063
## 68  1.660002e-04     70 0.03507605 0.03686243 0.0006038883
## 69  1.589071e-04     71 0.03491005 0.03676977 0.0006036293
## 70  1.551367e-04     72 0.03475115 0.03661088 0.0006032172
## 71  1.515608e-04     73 0.03459601 0.03643929 0.0006012496
## 72  1.461159e-04     74 0.03444445 0.03629750 0.0006006849
## 73  1.311620e-04     75 0.03429833 0.03603700 0.0005987670
## 74  1.294649e-04     76 0.03416717 0.03588286 0.0005979963
## 75  1.293837e-04     77 0.03403771 0.03585119 0.0005975625
## 76  1.282315e-04     78 0.03390832 0.03582295 0.0005969027
## 77  1.228990e-04     79 0.03378009 0.03566296 0.0005956697
## 78  1.228808e-04     80 0.03365719 0.03558829 0.0005943491
## 79  1.165335e-04     83 0.03328855 0.03537498 0.0005927021
## 80  1.148038e-04     84 0.03317202 0.03515673 0.0005911371
## 81  1.030627e-04     85 0.03305721 0.03499724 0.0005897277
## 82  1.023377e-04     86 0.03295415 0.03481631 0.0005884978
## 83  1.001605e-04     87 0.03285181 0.03475872 0.0005883899
## 84  9.863483e-05     88 0.03275165 0.03461843 0.0005877951
## 85  9.738345e-05     89 0.03265302 0.03453484 0.0005875253
## 86  9.723285e-05     90 0.03255563 0.03450710 0.0005870995
## 87  9.592221e-05     91 0.03245840 0.03450186 0.0005870466
## 88  9.522754e-05     92 0.03236248 0.03442264 0.0005864816
## 89  9.316224e-05     93 0.03226725 0.03436769 0.0005864436
## 90  9.282590e-05     94 0.03217409 0.03431830 0.0005862894
## 91  9.278577e-05     95 0.03208126 0.03431425 0.0005863251
## 92  9.255895e-05     96 0.03198848 0.03430023 0.0005861811
## 93  9.159710e-05     97 0.03189592 0.03422362 0.0005860079
## 94  8.946811e-05     98 0.03180432 0.03409151 0.0005847387
## 95  8.787822e-05     99 0.03171485 0.03401357 0.0005844216
## 96  8.767847e-05    100 0.03162697 0.03398024 0.0005842262
## 97  8.229727e-05    101 0.03153929 0.03387153 0.0005838517
## 98  8.182038e-05    102 0.03145700 0.03372571 0.0005836821
## 99  7.899661e-05    103 0.03137518 0.03360354 0.0005830723
## 100 7.860771e-05    104 0.03129618 0.03350421 0.0005822652
## 101 7.815595e-05    105 0.03121757 0.03349655 0.0005822470
## 102 7.784396e-05    106 0.03113942 0.03346652 0.0005821555
```

```
## 103 7.477239e-05    107 0.03106157 0.03335894 0.0005818341
## 104 7.277072e-05    108 0.03098680 0.03322757 0.0005806656
## 105 7.216759e-05    109 0.03091403 0.03315449 0.0005799720
## 106 7.160280e-05    110 0.03084186 0.03314802 0.0005799685
## 107 7.061479e-05    111 0.03077026 0.03309659 0.0005797633
## 108 6.944770e-05    112 0.03069964 0.03305018 0.0005797132
## 109 6.799173e-05    113 0.03063020 0.03300779 0.0005797052
## 110 6.536037e-05    114 0.03056221 0.03292746 0.0005791588
## 111 6.425304e-05    115 0.03049684 0.03290422 0.0005792821
## 112 6.127362e-05    116 0.03043259 0.03281238 0.0005787350
## 113 6.104617e-05    117 0.03037132 0.03272103 0.0005780112
## 114 6.101446e-05    118 0.03031027 0.03270863 0.0005779775
## 115 6.082340e-05    119 0.03024926 0.03270863 0.0005779775
## 116 6.041920e-05    120 0.03018843 0.03266225 0.0005776413
## 117 5.977735e-05    121 0.03012802 0.03264040 0.0005776046
## 118 5.866973e-05    122 0.03006824 0.03262213 0.0005775588
## 119 5.719734e-05    123 0.03000957 0.03251522 0.0005775470
## 120 5.584338e-05    124 0.02995237 0.03245257 0.0005766456
## 121 5.431397e-05    125 0.02989653 0.03236868 0.0005766859
## 122 5.388537e-05    126 0.02984221 0.03232405 0.0005762053
## 123 5.370637e-05    127 0.02978833 0.03232190 0.0005761523
## 124 5.209662e-05    128 0.02973462 0.03229686 0.0005760984
## 125 5.142471e-05    129 0.02968252 0.03223578 0.0005755245
## 126 5.097416e-05    130 0.02963110 0.03220611 0.0005753075
## 127 5.004255e-05    131 0.02958013 0.03217305 0.0005755400
## 128 4.957652e-05    132 0.02953008 0.03214527 0.0005752423
## 129 4.908334e-05    133 0.02948051 0.03210842 0.0005750871
## 130 4.874270e-05    134 0.02943142 0.03210732 0.0005750861
## 131 4.867408e-05    135 0.02938268 0.03209131 0.0005750570
## 132 4.835055e-05    136 0.02933401 0.03208697 0.0005753835
## 133 4.765415e-05    137 0.02928566 0.03202259 0.0005750322
## 134 4.763924e-05    138 0.02923800 0.03197948 0.0005748691
## 135 4.733672e-05    139 0.02919036 0.03195954 0.0005747871
## 136 4.578921e-05    140 0.02914303 0.03190456 0.0005747080
## 137 4.537540e-05    141 0.02909724 0.03188225 0.0005747726
## 138 4.493991e-05    142 0.02905186 0.03184746 0.0005747121
## 139 4.490278e-05    143 0.02900692 0.03182931 0.0005745585
## 140 4.362800e-05    144 0.02896202 0.03179472 0.0005744951
## 141 4.341470e-05    145 0.02891839 0.03176056 0.0005744573
## 142 4.253115e-05    147 0.02883156 0.03171426 0.0005743496
## 143 4.247114e-05    148 0.02878903 0.03168379 0.0005743355
## 144 4.193812e-05    149 0.02874656 0.03164288 0.0005744321
## 145 4.111105e-05    150 0.02870462 0.03159148 0.0005743506
## 146 4.083397e-05    151 0.02866351 0.03154841 0.0005740017
## 147 4.009935e-05    152 0.02862268 0.03150304 0.0005739340
## 148 3.970545e-05    154 0.02854248 0.03144930 0.0005740044
## 149 3.948265e-05    155 0.02850277 0.03142962 0.0005739953
## 150 3.733818e-05    156 0.02846329 0.03129767 0.0005733983
## 151 3.701214e-05    157 0.02842595 0.03121927 0.0005732820
## 152 3.698288e-05    158 0.02838894 0.03122181 0.0005736642
```

```
## 153 3.679665e-05    159 0.02835196 0.03121563 0.0005736584
## 154 3.675895e-05    160 0.02831516 0.03119844 0.0005736189
## 155 3.655037e-05    161 0.02827840 0.03118900 0.0005736097
## 156 3.620663e-05    162 0.02824185 0.03114625 0.0005735910
## 157 3.485543e-05    163 0.02820564 0.03109645 0.0005735241
## 158 3.457082e-05    164 0.02817079 0.03104263 0.0005734470
## 159 3.419363e-05    165 0.02813622 0.03101499 0.0005731074
## 160 3.301284e-05    166 0.02810202 0.03093935 0.0005730104
## 161 3.233906e-05    167 0.02806901 0.03088167 0.0005731459
## 162 3.206395e-05    168 0.02803667 0.03086873 0.0005733310
## 163 3.060887e-05    169 0.02800461 0.03084614 0.0005732885
## 164 3.054593e-05    170 0.02797400 0.03083126 0.0005732886
## 165 2.849167e-05    171 0.02794345 0.03080007 0.0005732848
## 166 2.819010e-05    172 0.02791496 0.03073320 0.0005731548
## 167 2.785897e-05    173 0.02788677 0.03072724 0.0005731501
## 168 2.697730e-05    174 0.02785891 0.03069809 0.0005731670
## 169 2.692539e-05    175 0.02783193 0.03067308 0.0005733908
## 170 2.657090e-05    176 0.02780501 0.03066849 0.0005733874
## 171 2.652532e-05    177 0.02777844 0.03065074 0.0005733864
## 172 2.638953e-05    178 0.02775191 0.03064651 0.0005734024
## 173 2.638608e-05    179 0.02772552 0.03063651 0.0005733634
## 174 2.544172e-05    180 0.02769914 0.03060193 0.0005732565
## 175 2.496005e-05    181 0.02767370 0.03053733 0.0005731174
## 176 2.484979e-05    182 0.02764874 0.03052701 0.0005731444
## 177 2.481766e-05    183 0.02762389 0.03052301 0.0005731451
## 178 2.463309e-05    184 0.02759907 0.03052110 0.0005731587
## 179 2.450909e-05    185 0.02757444 0.03050758 0.0005731520
## 180 2.418173e-05    186 0.02754993 0.03049113 0.0005731606
## 181 2.409481e-05    187 0.02752574 0.03045769 0.0005730734
## 182 2.344556e-05    188 0.02750165 0.03043380 0.0005730152
## 183 2.291530e-05    189 0.02747820 0.03040836 0.0005731537
## 184 2.225707e-05    190 0.02745529 0.03036156 0.0005728650
## 185 2.158568e-05    191 0.02743303 0.03033004 0.0005728358
## 186 2.153589e-05    192 0.02741145 0.03031500 0.0005728160
## 187 2.150643e-05    193 0.02738991 0.03031467 0.0005728234
## 188 2.074862e-05    194 0.02736840 0.03028544 0.0005727761
## 189 2.067639e-05    195 0.02734766 0.03026359 0.0005727219
## 190 2.061707e-05    196 0.02732698 0.03025420 0.0005727140
## 191 2.019546e-05    197 0.02730636 0.03023127 0.0005727193
## 192 2.004852e-05    198 0.02728617 0.03021456 0.0005726886
## 193 1.945513e-05    199 0.02726612 0.03019006 0.0005726324
## 194 1.928330e-05    200 0.02724666 0.03017996 0.0005727231
## 195 1.896996e-05    201 0.02722738 0.03016836 0.0005727150
## 196 1.859715e-05    202 0.02720841 0.03015131 0.0005726913
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## 465 5.532338e-07    489 0.02577933 0.02901578 0.0005730515
## 466 5.507121e-07    490 0.02577877 0.02901528 0.0005730519
## 467 5.305677e-07    491 0.02577822 0.02901468 0.0005730524
## 468 5.258540e-07    492 0.02577769 0.02901392 0.0005730527
## 469 5.176442e-07    493 0.02577717 0.02901365 0.0005730529
## 470 5.141519e-07    497 0.02577495 0.02901362 0.0005730530
## 471 4.955436e-07    498 0.02577444 0.02901255 0.0005730534
## 472 4.920332e-07    499 0.02577394 0.02901180 0.0005730540
## 473 4.907174e-07    500 0.02577345 0.02901163 0.0005730542
## 474 4.826913e-07    502 0.02577247 0.02901151 0.0005730543
## 475 4.651042e-07    503 0.02577198 0.02901067 0.0005730551
## 476 4.611899e-07    504 0.02577152 0.02901040 0.0005730552
## 477 4.424597e-07    506 0.02577060 0.02900974 0.0005730557
## 478 4.380939e-07    507 0.02577015 0.02900913 0.0005730563
## 479 4.373741e-07    508 0.02576972 0.02900913 0.0005730563
## 480 4.275044e-07    509 0.02576928 0.02900894 0.0005730565
## 481 4.224906e-07    510 0.02576885 0.02900829 0.0005730571
## 482 4.218445e-07    511 0.02576843 0.02900824 0.0005730572
## 483 4.004741e-07    512 0.02576801 0.02900746 0.0005730578
## 484 3.996750e-07    513 0.02576761 0.02900626 0.0005730590
## 485 3.974896e-07    514 0.02576721 0.02900618 0.0005730590
## 486 3.906966e-07    516 0.02576641 0.02900607 0.0005730591
## 487 3.818977e-07    517 0.02576602 0.02900530 0.0005730601
## 488 3.705889e-07    518 0.02576564 0.02900467 0.0005730606
## 489 3.636678e-07    519 0.02576527 0.02900415 0.0005730610
## 490 3.519816e-07    520 0.02576490 0.02900381 0.0005730613
## 491 3.417130e-07    521 0.02576455 0.02900315 0.0005730619
## 492 3.403755e-07    522 0.02576421 0.02900306 0.0005730619
## 493 3.347550e-07    523 0.02576387 0.02900293 0.0005730620
## 494 3.344891e-07    524 0.02576354 0.02900281 0.0005730621
## 495 3.322664e-07    525 0.02576320 0.02900282 0.0005730621
## 496 3.264275e-07    526 0.02576287 0.02900271 0.0005730622
## 497 3.261576e-07    527 0.02576254 0.02900256 0.0005730623
## 498 3.188002e-07    528 0.02576222 0.02900250 0.0005730624
## 499 3.061858e-07    529 0.02576190 0.02900186 0.0005730629
## 500 3.020125e-07    530 0.02576159 0.02900163 0.0005730632
## 501 3.003704e-07    531 0.02576129 0.02900155 0.0005730632
## 502 2.986556e-07    532 0.02576099 0.02900154 0.0005730632
```

```
## 503 2.837937e-07    533 0.02576069 0.02900099 0.0005730637
## 504 2.755890e-07    534 0.02576041 0.02900013 0.0005730644
## 505 2.734581e-07    535 0.02576013 0.02899979 0.0005730647
## 506 2.728056e-07    536 0.02575986 0.02899977 0.0005730647
## 507 2.673842e-07    537 0.02575958 0.02899970 0.0005730648
## 508 2.651059e-07    538 0.02575932 0.02899933 0.0005730651
## 509 2.633776e-07    539 0.02575905 0.02899931 0.0005730652
## 510 2.611412e-07    540 0.02575879 0.02899928 0.0005730652
## 511 2.588331e-07    541 0.02575853 0.02899932 0.0005730651
## 512 2.520916e-07    542 0.02575827 0.02899928 0.0005730652
## 513 2.483592e-07    543 0.02575802 0.02899916 0.0005730653
## 514 2.453175e-07    544 0.02575777 0.02899897 0.0005730655
## 515 2.447042e-07    546 0.02575728 0.02899893 0.0005730655
## 516 2.444947e-07    547 0.02575703 0.02899893 0.0005730655
## 517 2.379747e-07    548 0.02575679 0.02899886 0.0005730656
## 518 2.350813e-07    549 0.02575655 0.02899841 0.0005730658
## 519 2.348044e-07    550 0.02575632 0.02899823 0.0005730660
## 520 2.315343e-07    551 0.02575608 0.02899816 0.0005730661
## 521 2.291409e-07    552 0.02575585 0.02899803 0.0005730662
## 522 2.237500e-07    553 0.02575562 0.02899786 0.0005730663
## 523 2.192458e-07    554 0.02575540 0.02899771 0.0005730665
## 524 2.044881e-07    555 0.02575518 0.02899757 0.0005730666
## 525 2.042590e-07    556 0.02575497 0.02899719 0.0005730669
## 526 1.962127e-07    557 0.02575477 0.02899715 0.0005730670
## 527 1.888082e-07    558 0.02575457 0.02899706 0.0005730670
## 528 1.886300e-07    560 0.02575419 0.02899698 0.0005730671
## 529 1.827287e-07    561 0.02575401 0.02899689 0.0005730672
## 530 1.800639e-07    562 0.02575382 0.02899684 0.0005730672
## 531 1.708380e-07    563 0.02575364 0.02899675 0.0005730672
## 532 1.658214e-07    564 0.02575347 0.02899663 0.0005730673
## 533 1.627412e-07    565 0.02575331 0.02899660 0.0005730674
## 534 1.470097e-07    566 0.02575314 0.02899657 0.0005730674
## 535 1.459686e-07    567 0.02575300 0.02899659 0.0005730674
## 536 1.425552e-07    568 0.02575285 0.02899656 0.0005730674
## 537 1.311014e-07    569 0.02575271 0.02899639 0.0005730676
## 538 1.271096e-07    570 0.02575258 0.02899631 0.0005730676
## 539 1.223215e-07    571 0.02575245 0.02899620 0.0005730677
## 540 1.200096e-07    572 0.02575233 0.02899616 0.0005730678
## 541 1.153567e-07    573 0.02575221 0.02899591 0.0005730680
## 542 1.146300e-07    574 0.02575209 0.02899583 0.0005730681
## 543 1.115429e-07    575 0.02575198 0.02899578 0.0005730681
## 544 9.376439e-08    576 0.02575187 0.02899564 0.0005730683
## 545 8.524972e-08    577 0.02575177 0.02899544 0.0005730684
## 546 8.402042e-08    578 0.02575169 0.02899539 0.0005730685
## 547 8.143882e-08    579 0.02575160 0.02899535 0.0005730685
## 548 7.867458e-08    580 0.02575152 0.02899537 0.0005730685
## 549 6.551576e-08    581 0.02575144 0.02899526 0.0005730686
## 550 5.540776e-08    582 0.02575138 0.02899515 0.0005730687
## 551 5.501848e-08    583 0.02575132 0.02899519 0.0005730687
## 552 5.477943e-08    584 0.02575127 0.02899519 0.0005730687
```

```
## 553 5.215047e-08    585 0.02575121 0.02899519 0.0005730687
## 554 2.611955e-08    586 0.02575116 0.02899518 0.0005730687
## 555 0.000000e+00    587 0.02575113 0.02899519 0.0005730687
```

```
TREE <- rpart(price ~ ., data = DIAMONDS, cp = 1.005400e-02, minbucket = 50)
visualize_model(TREE)
```


```
# Make special datasets for examining partial dependence plots, interactions,
and breakdown plots
TRAIN.PREDICTORS <- DIAMONDS
TRAIN.PREDICTORS$price <- NULL
TRAIN.TARGET <- DIAMONDS$price

# Estimating generalization error (build on a 20% training set and evaluate o
n an 80% holdout sample)
# Since data is so big
# Normally this is done with K-fold crossvalidation and caret, but this is fa
ster!
# Drawback:  only one guess of the generalization error!
DATA <- DIAMONDS
y <- "price"
set.seed(617904); train.rows <- sample(1:nrow(DATA), 0.2 * nrow(DATA))
MODEL <- randomForest(formula(paste(y, "~.")), data = DATA[train.rows, ])
```

```r
# Estimated RMSE on holdout sample (one measure of generalization error)
postResample(predict(MODEL, newdata = DATA[-train.rows, ]), DATA[-train.rows, y])
```

```
##        RMSE    Rsquared         MAE
## 647.1698304   0.9742289 327.4005341
```

```r
# RMSE on holdout if predicted everything to be average value (naive model)
postResample(mean(DATA[train.rows, y]), DATA[-train.rows, y])
```

```
##     RMSE Rsquared      MAE
## 3994.086       NA 3017.672
```

```r
# Building an explainer with a random forest
library(randomForest)
FOREST <- randomForest(formula(paste(y, "~.")), data = DATA[train.rows, ])
summarize_tree(FOREST)
```

```
## $importance
##           y       carat           x           z     clarity       color
## 57317443092 43493594961 30396518072 22608440004  8830195974  4554207554
##       depth       table         cut
##  1063276663   792436512   718849640
```

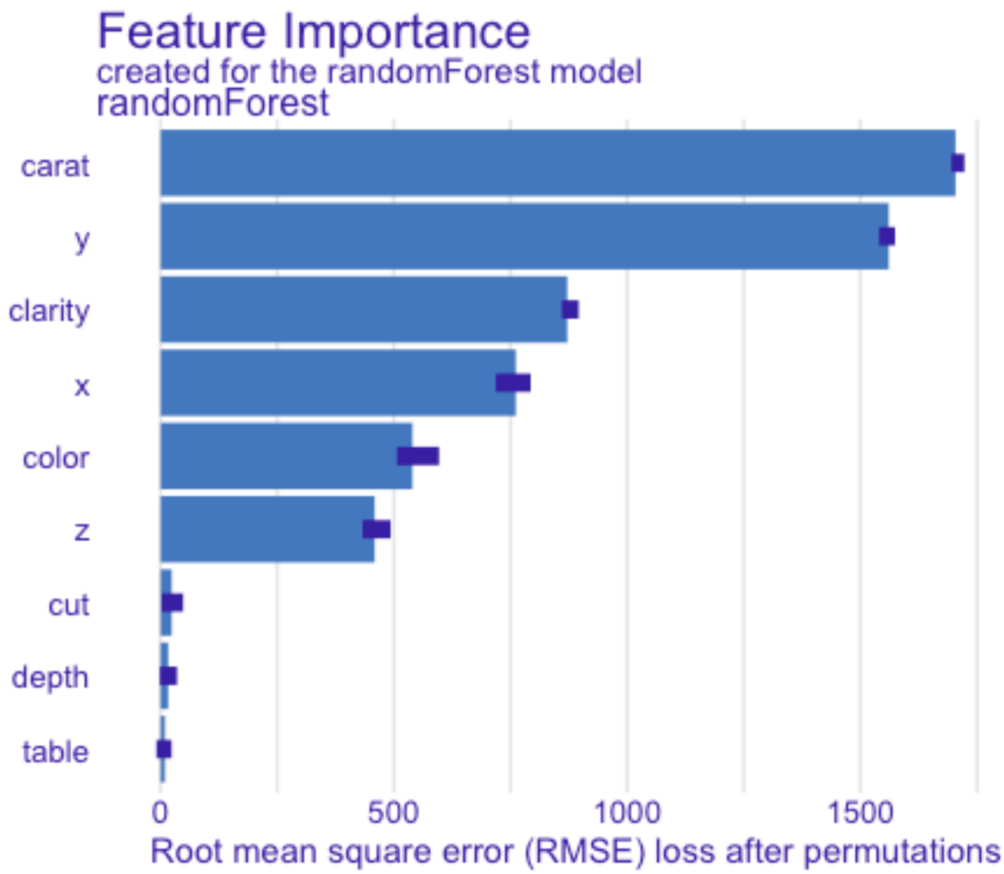
```r
model_explainer <- DALEX::explain(FOREST, data = TRAIN.PREDICTORS,  y = TRAIN
.TARGET)
```

```
## Preparation of a new explainer is initiated
##   -> model label       :  randomForest  (  default  )
##   -> data              :  53940  rows  9  cols
##   -> target variable   :  53940  values
##   -> predict function  :  yhat.randomForest  will be used (  default  )
##   -> predicted values  :  No value for predict function target column. (
default  )
##   -> model_info        :  package randomForest , ver. 4.7.1.2 , task regre
ssion (  default  )
##   -> predicted values  :  numerical, min =  384.8449 , mean =  3921.51 , m
ax =  18014.24
##   -> residual function :  difference between y and yhat (  default  )
##   -> residuals         :  numerical, min =  -6309.706 , mean =  11.29 , ma
x =  10465.68
##   A new explainer has been created!
```

```r
model_vi <- model_parts(model_explainer, type = "difference")
plot(model_vi)
```

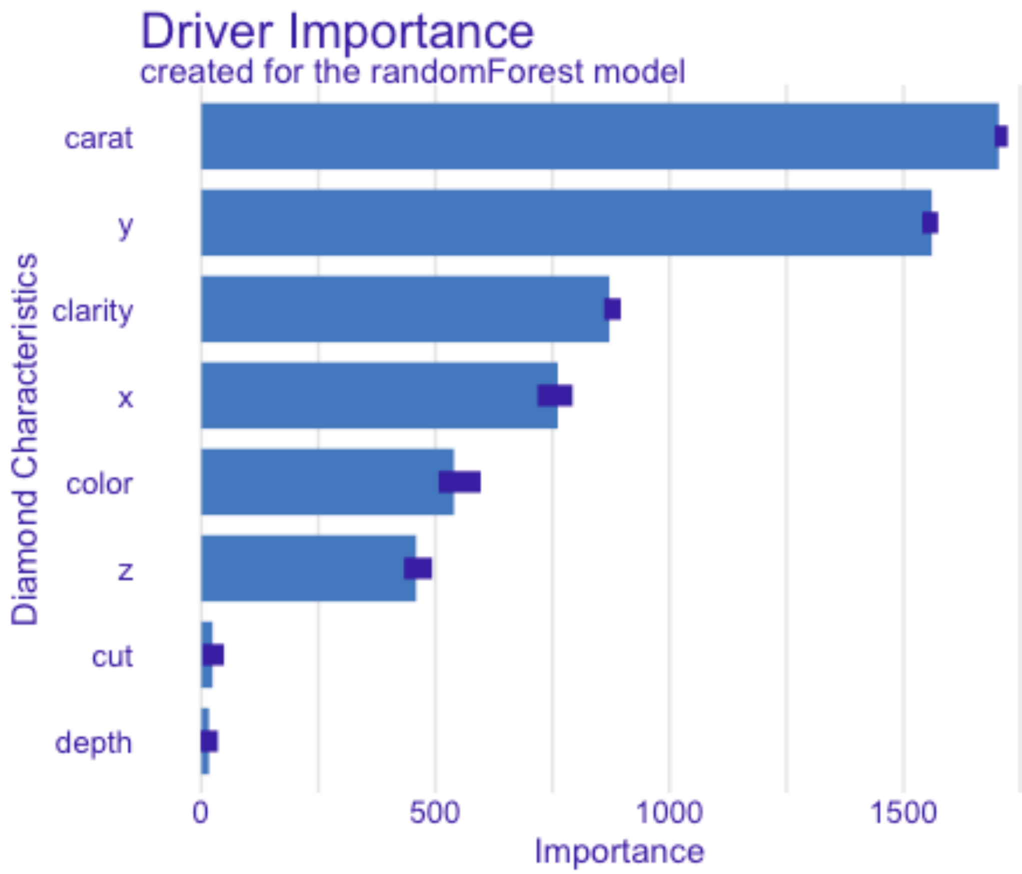
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## ℹ The deprecated feature was likely used in the ingredients package.
##   Please report the issue at
##   <https://github.com/ModelOriented/ingredients/issues>.
## This warning is displayed once every 8 hours.
```

```
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



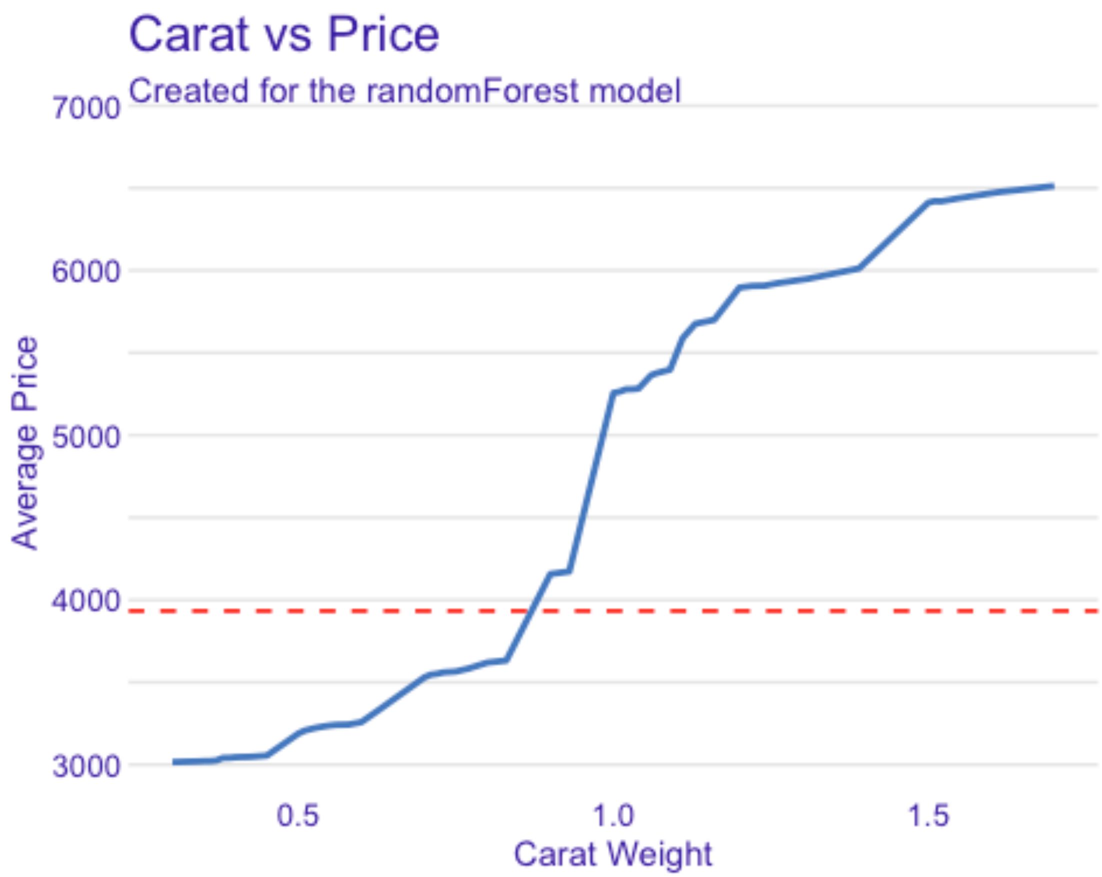

```
model_vi

##           variable mean_dropout_loss        label
## 1  _full_model_          0.000000 randomForest
## 2         table          9.583449 randomForest
## 3         depth         16.378242 randomForest
## 4           cut         23.967071 randomForest
## 5             z        458.838143 randomForest
## 6         color        539.816540 randomForest
## 7             x        761.448278 randomForest
## 8       clarity        871.740926 randomForest
## 9             y       1560.517549 randomForest
## 10        carat       1703.574357 randomForest
## 11   _baseline_       4957.611843 randomForest

# Polish it up a bit (you'll want to tweak max_vars)
plot(model_vi, max_vars = 8) +
  labs(
    title = "Driver Importance",
    x = "Diamond Characteristics",
    y = "Importance"
```

```r
  ) +
  theme(strip.text.x = element_blank())
```



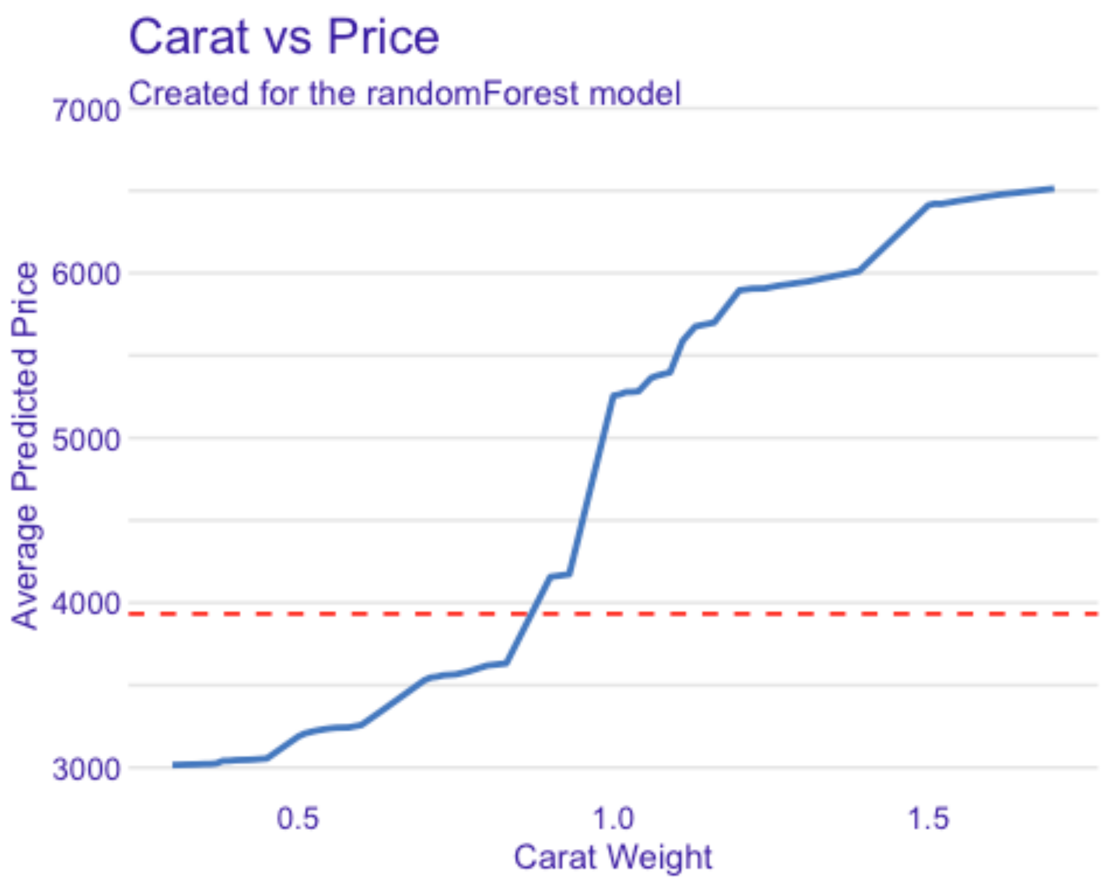

```r
numerics <- names(DATA)[which(unlist(lapply(DATA, function(x) tail(class(x),
1))) %in% c("numeric", "integer"))]
numerics <- setdiff(numerics, y)
cats <- names(DATA)[which(unlist(lapply(DATA, function(x) tail(class(x), 1)))
%in% c("factor", "character"))]
cats <- setdiff(cats, y) #Take out name of y variable


# Partial dependence plots for numeric variables (may take a while)
# Ignore "has more than 201 unique values" messages
set.seed(617904); pd_numerics <- model_profile(model_explainer, type = "parti
al", variables = numerics)


# Example with one variable (carat, older template version – kept but cleaned
)
plot(pd_numerics, variables = "carat") +
  labs(
    title = "Carat vs Price",
    x = "Carat Weight",
```

```
    y = "Average Price"
  ) +
  lims(x = as.numeric(quantile(DATA$carat, c(.05, .95)))) +
  geom_hline(yintercept = mean(TRAIN.TARGET), linetype = "dashed", color = "r
ed", linewidth = 0.6) +
  theme(strip.text.x = element_blank())

## Warning: Removed 8 rows containing missing values or values outside the sc
ale range
## (`geom_line()`).
```



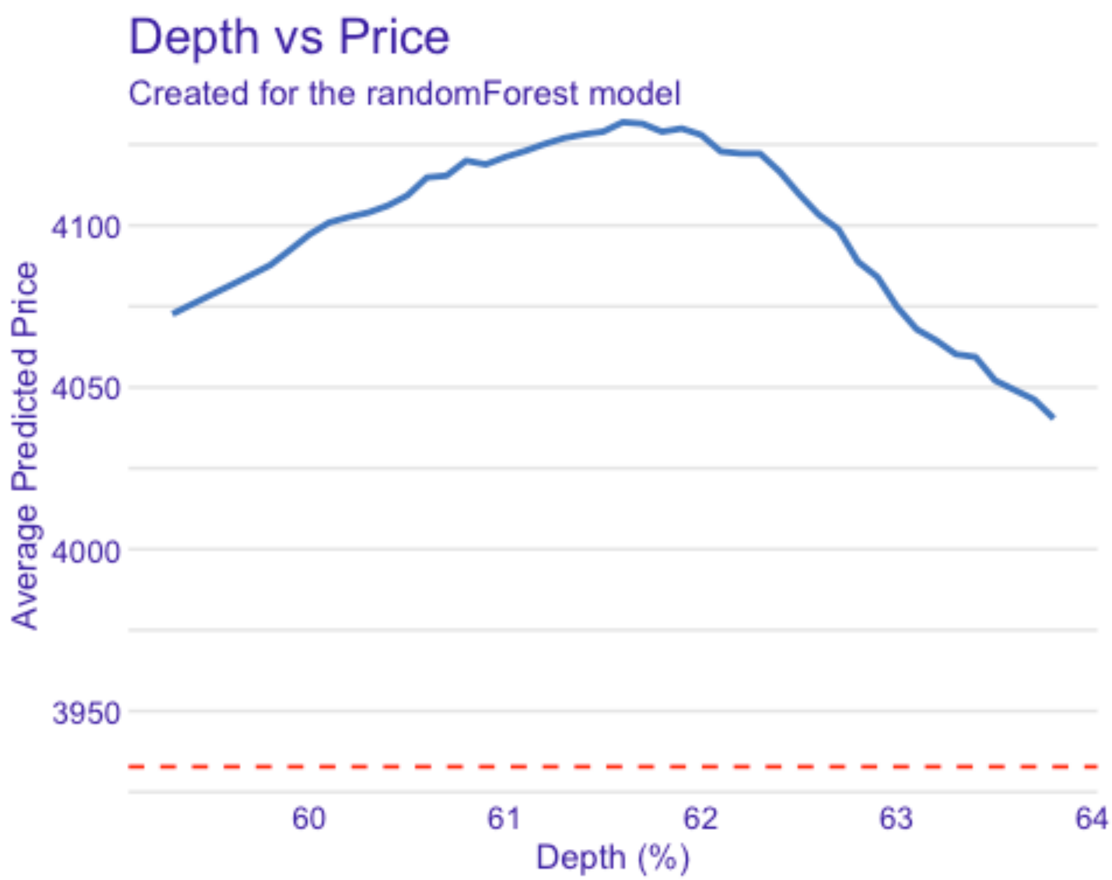

```
# Tweaked PDP for carat
plot(pd_numerics, variables = "carat") +
  labs(
    title = "Carat vs Price",
    x = "Carat Weight",
    y = "Average Predicted Price"
  ) +
  lims(x = as.numeric(quantile(DATA$carat, c(.05, .95)))) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(strip.text.x = element_blank())
```

```
## Warning: Removed 8 rows containing missing values or values outside the sc
ale range
## (`geom_line()`).
```



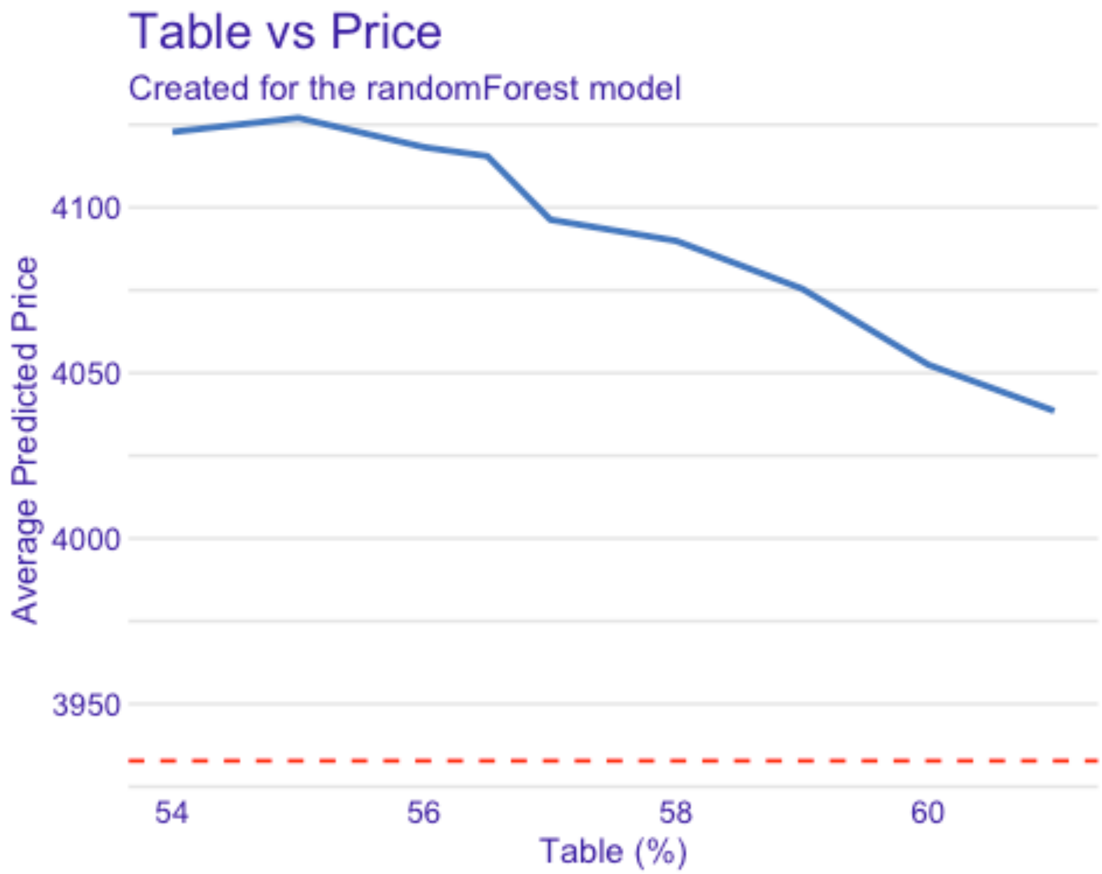

```r
# Tweaked PDP for depth
plot(pd_numerics, variables = "depth") +
  labs(
    title = "Depth vs Price",
    x = "Depth (%)",
    y = "Average Predicted Price"
  ) +
  lims(x = as.numeric(quantile(DATA$depth, c(.05, .95)))) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(strip.text.x = element_blank())
```

```
## Warning: Removed 10 rows containing missing values or values outside the s
cale range
## (`geom_line()`).
```




```
# Tweaked PDP for table
plot(pd_numerics, variables = "table") +
  labs(
    title = "Table vs Price",
    x = "Table (%)",
    y = "Average Predicted Price"
  ) +
  lims(x = as.numeric(quantile(DATA$table, c(.05, .95)))) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(strip.text.x = element_blank())
```

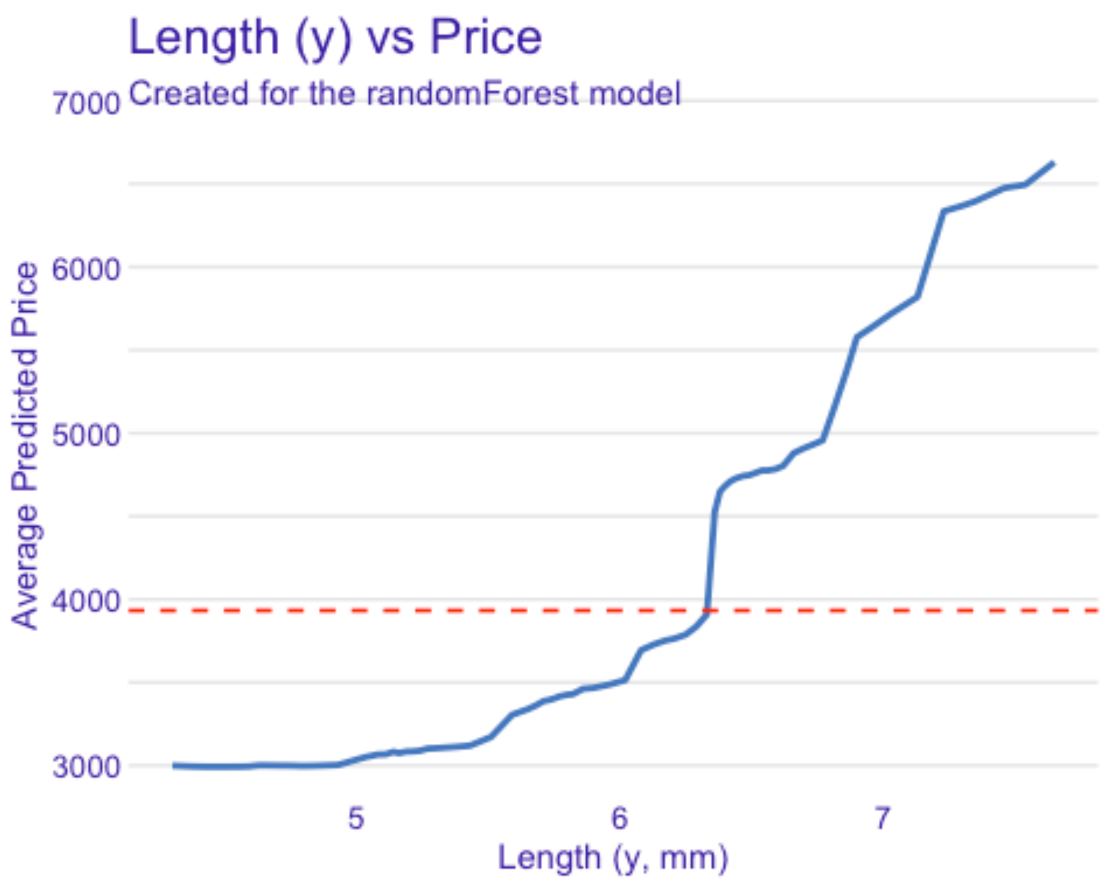
```
## Warning: Removed 6 rows containing missing values or values outside the sc
ale range
## (`geom_line()`).
```




```
# Tweaked PDP for x
plot(pd_numerics, variables = "x") +
  labs(
    title = "Width (x) vs Price",
    x = "Width (x, mm)",
    y = "Average Predicted Price"
  ) +
  lims(x = as.numeric(quantile(DATA$x, c(.05, .95)))) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(strip.text.x = element_blank())
```

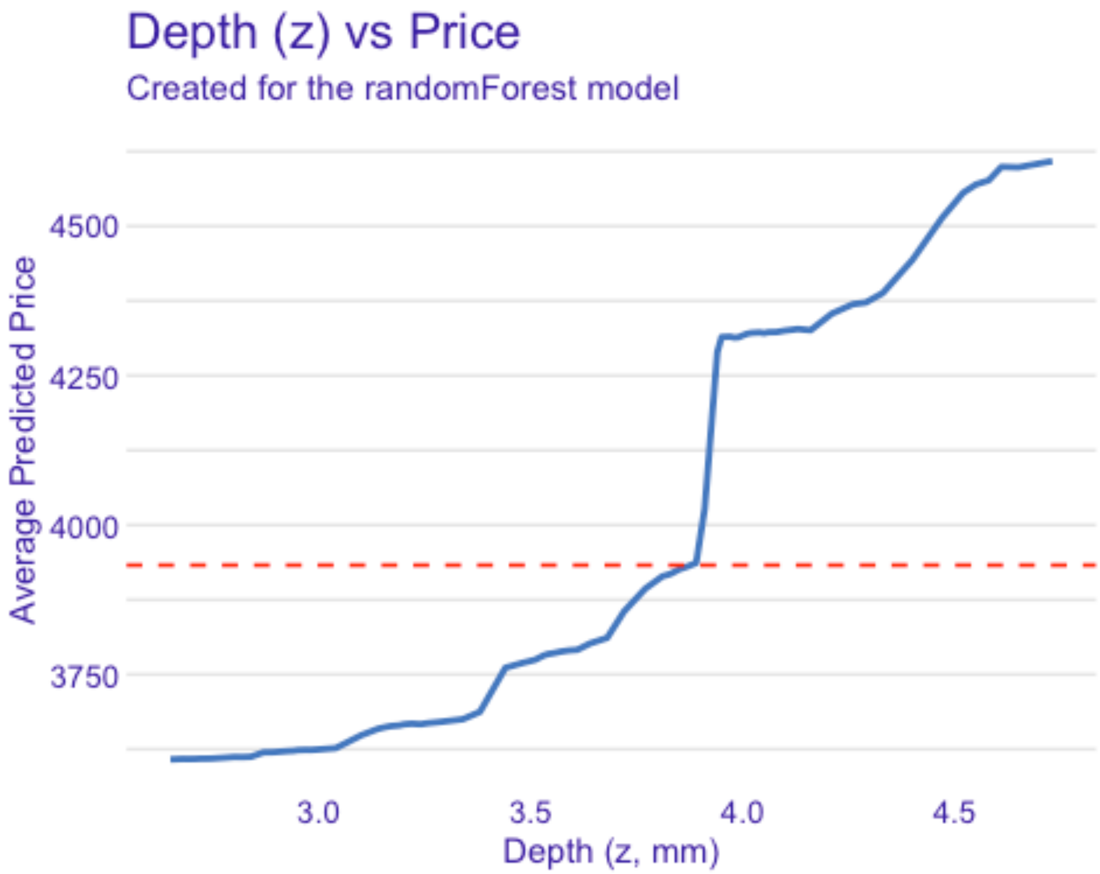
```
## Warning: Removed 10 rows containing missing values or values outside the s
cale range
## (`geom_line()`).
```


```
# Tweaked PDP for y
plot(pd_numerics, variables = "y") +
  labs(
    title = "Length (y) vs Price",
    x = "Length (y, mm)",
    y = "Average Predicted Price"
  ) +
  lims(x = as.numeric(quantile(DATA$y, c(.05, .95)))) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(strip.text.x = element_blank())

## Warning: Removed 10 rows containing missing values or values outside the s
cale range
## (`geom_line()`).
```




```
# Tweaked PDP for z
plot(pd_numerics, variables = "z") +
  labs(
    title = "Depth (z) vs Price",
    x = "Depth (z, mm)",
    y = "Average Predicted Price"
  ) +
  lims(x = as.numeric(quantile(DATA$z, c(.05, .95)))) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(strip.text.x = element_blank())
```

```
## Warning: Removed 10 rows containing missing values or values outside the s
cale range
## (`geom_line()`).
```

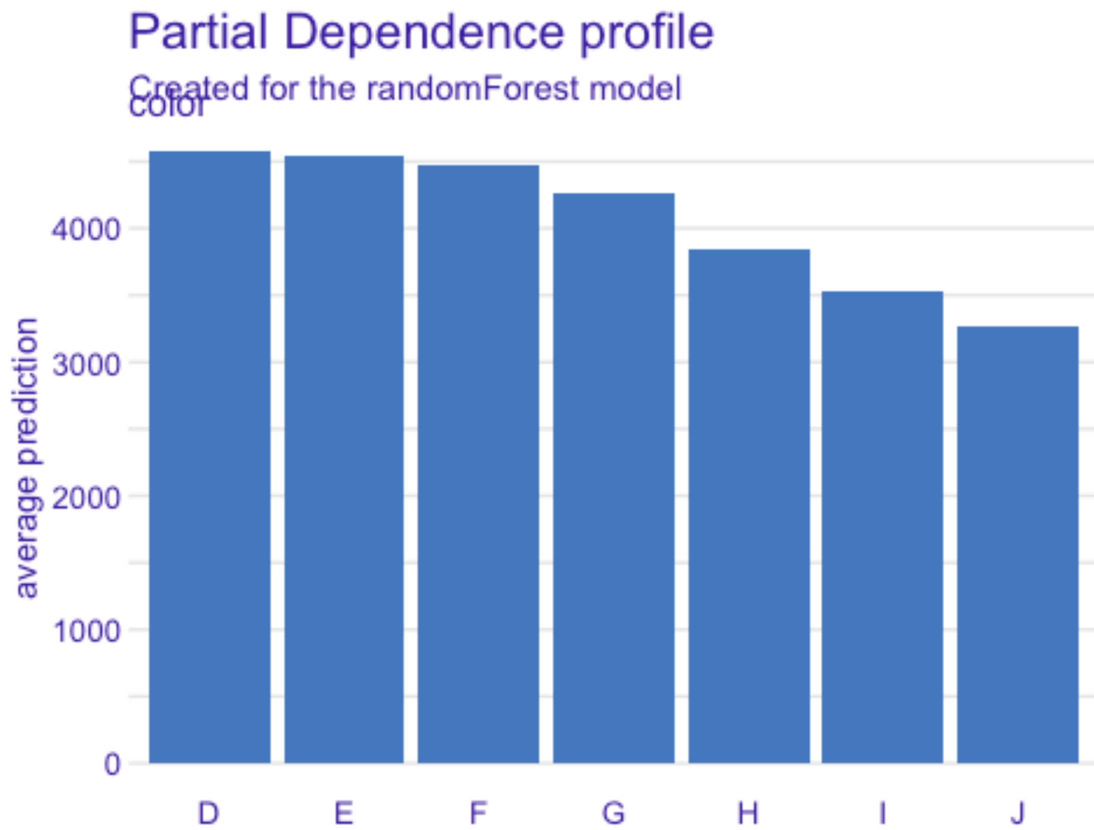



```
# Categorical drivers (ignore 'variable_type' changed to 'categorical' due to
lack of numerical variables)
set.seed(617904); pd_cats <- model_profile(model_explainer, type = "partial",
variables = cats)
```

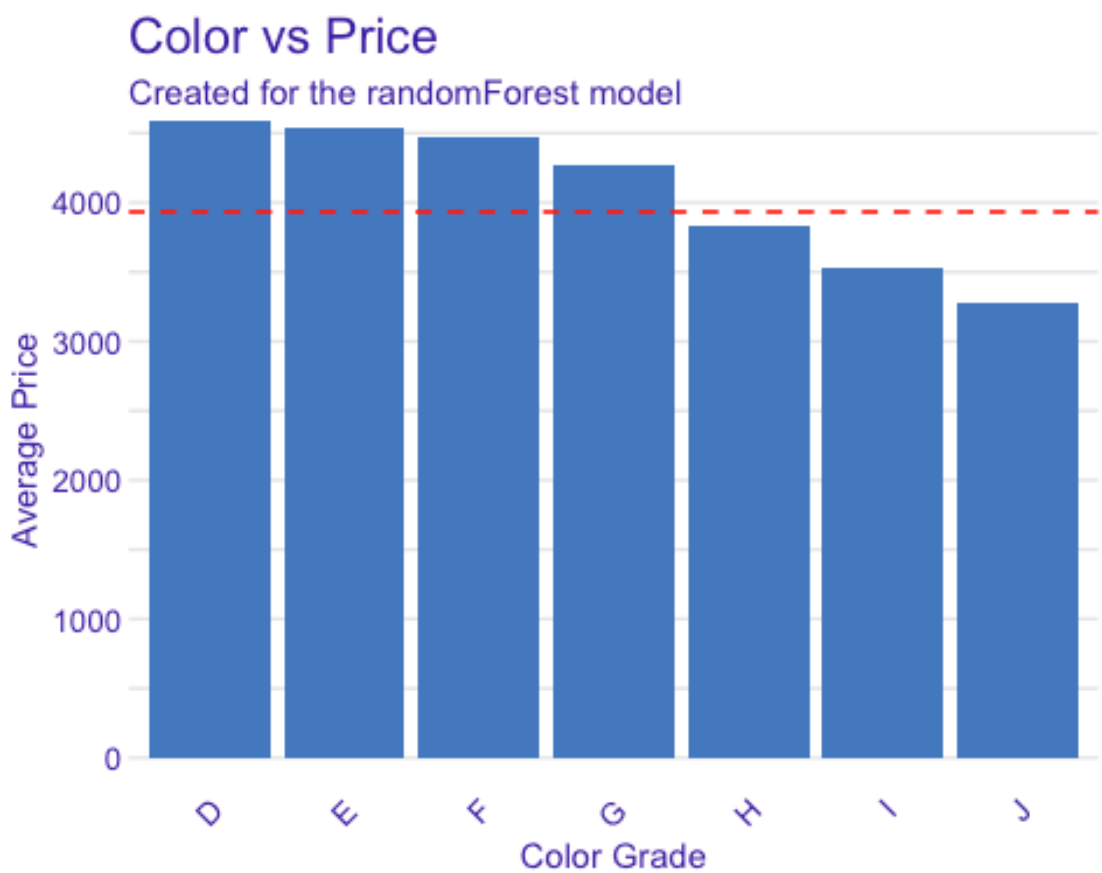
```
## 'variable_type' changed to 'categorical' due to lack of numerical variable
s.
```

```
# Default presentation
plot(pd_cats, variables = "color")
```

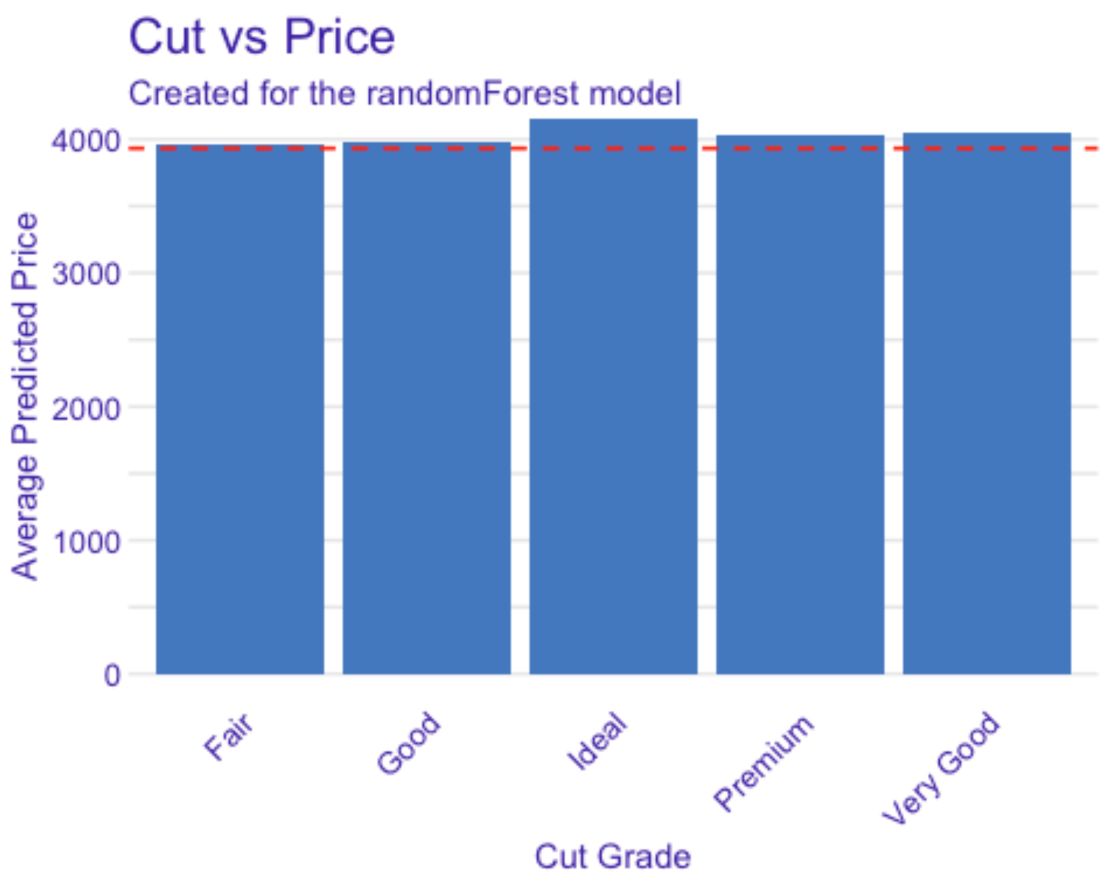
```
## Warning in geom_col(aes(y = `_yhat_`), size = size, alpha = alpha, fill =
## color, : Ignoring unknown parameters: `size`
```




```
# Tweaked and better for color (original example)
plot(pd_cats, variables = "color") +
  labs(
    title = "Color vs Price",
    x = "Color Grade",
    y = "Average Price"
  ) +
  geom_hline(yintercept = mean(TRAIN.TARGET), linetype = "dashed", color = "r
ed", linewidth = 0.6) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 10)) +
  theme(strip.text.x = element_blank())

## Warning in geom_col(aes(y = `_yhat_`), size = size, alpha = alpha, fill =
## color, : Ignoring unknown parameters: `size`
```

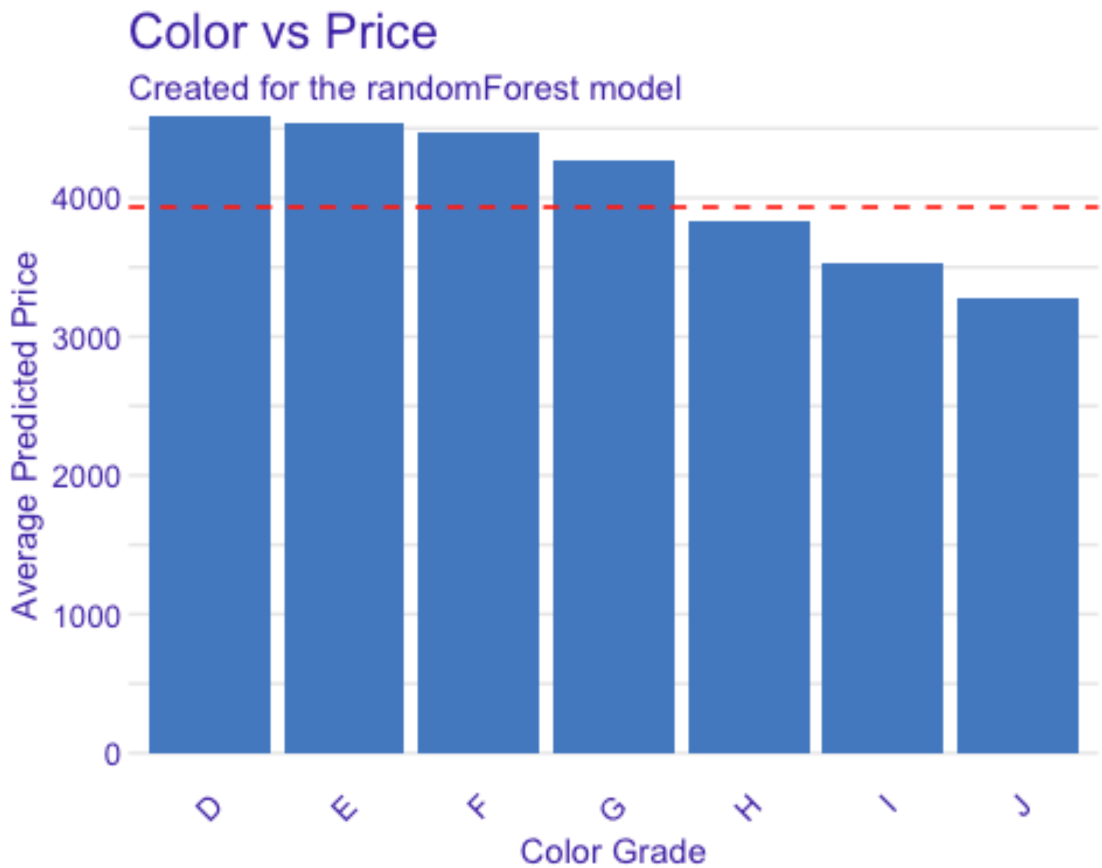



```
# Tweaked PDP for Cut
plot(pd_cats, variables = "cut") +
  labs(
    title = "Cut vs Price",
    x = "Cut Grade",
    y = "Average Predicted Price"
  ) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 10)) +
  theme(strip.text.x = element_blank())

## Warning in geom_col(aes(y = `_yhat_`), size = size, alpha = alpha, fill =
## color, : Ignoring unknown parameters: `size`
```



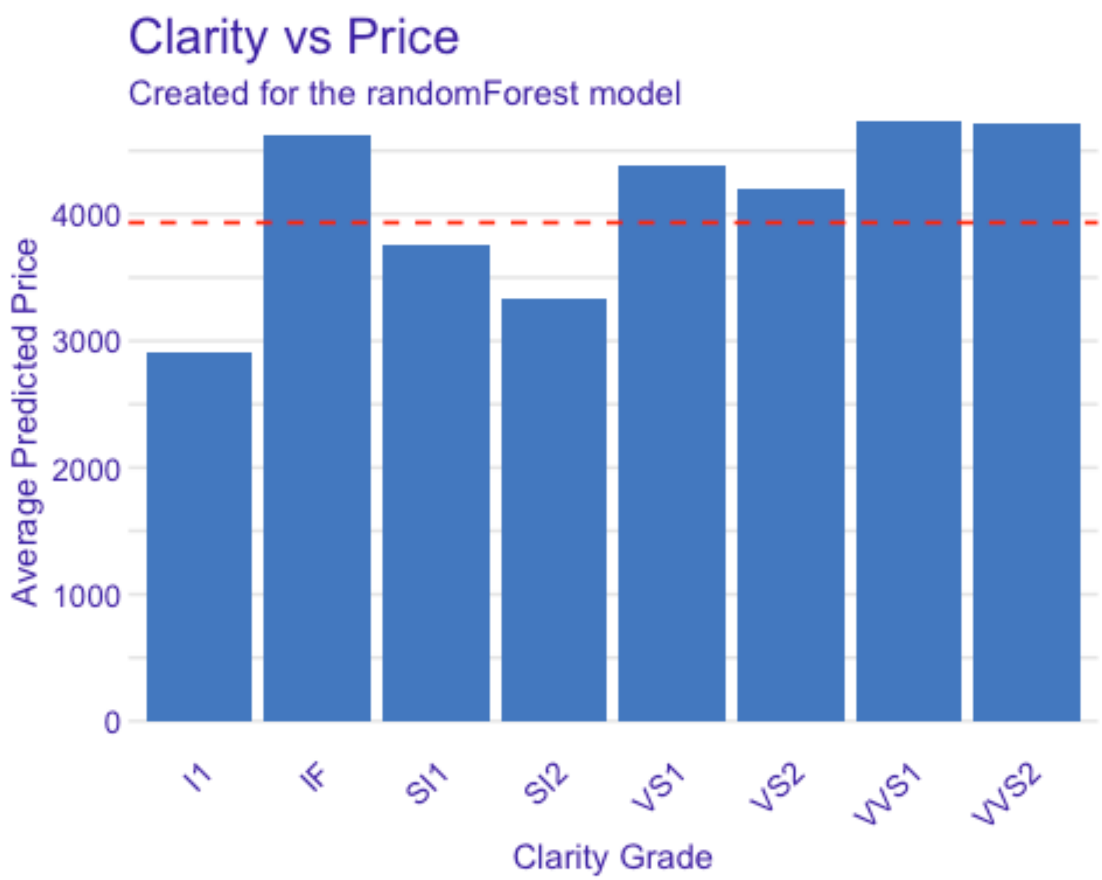

```
# Tweaked PDP for Color
plot(pd_cats, variables = "color") +
  labs(
    title = "Color vs Price",
    x = "Color Grade",
    y = "Average Predicted Price"
  ) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 10)) +
  theme(strip.text.x = element_blank())
```

```
## Warning in geom_col(aes(y = `_yhat_`), size = size, alpha = alpha, fill =
## color, : Ignoring unknown parameters: `size`
```

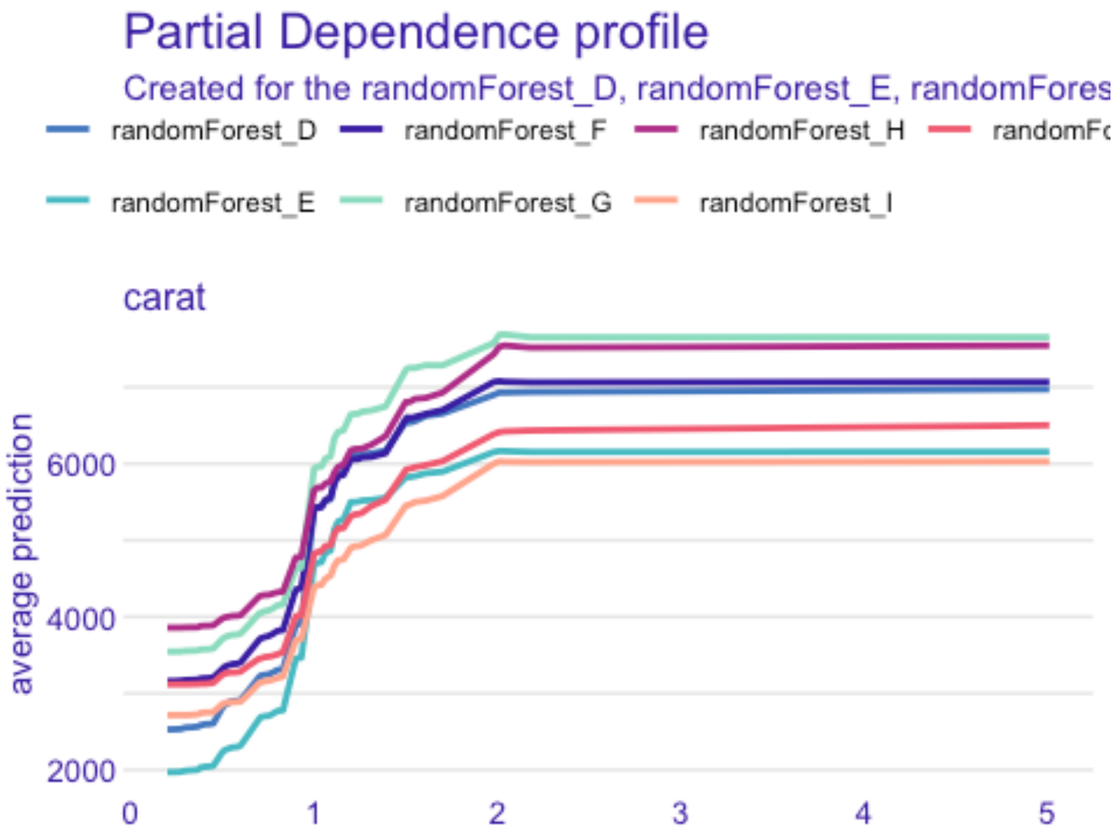



```
# Tweaked PDP for Clarity
plot(pd_cats, variables = "clarity") +
  labs(
    title = "Clarity vs Price",
    x = "Clarity Grade",
    y = "Average Predicted Price"
  ) +
  geom_hline(yintercept = mean(TRAIN.TARGET),
             linetype = "dashed", color = "red", linewidth = 0.6) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 10)) +
  theme(strip.text.x = element_blank())
```

```
## Warning in geom_col(aes(y = `_yhat_`), size = size, alpha = alpha, fill =
## color, : Ignoring unknown parameters: `size`
```

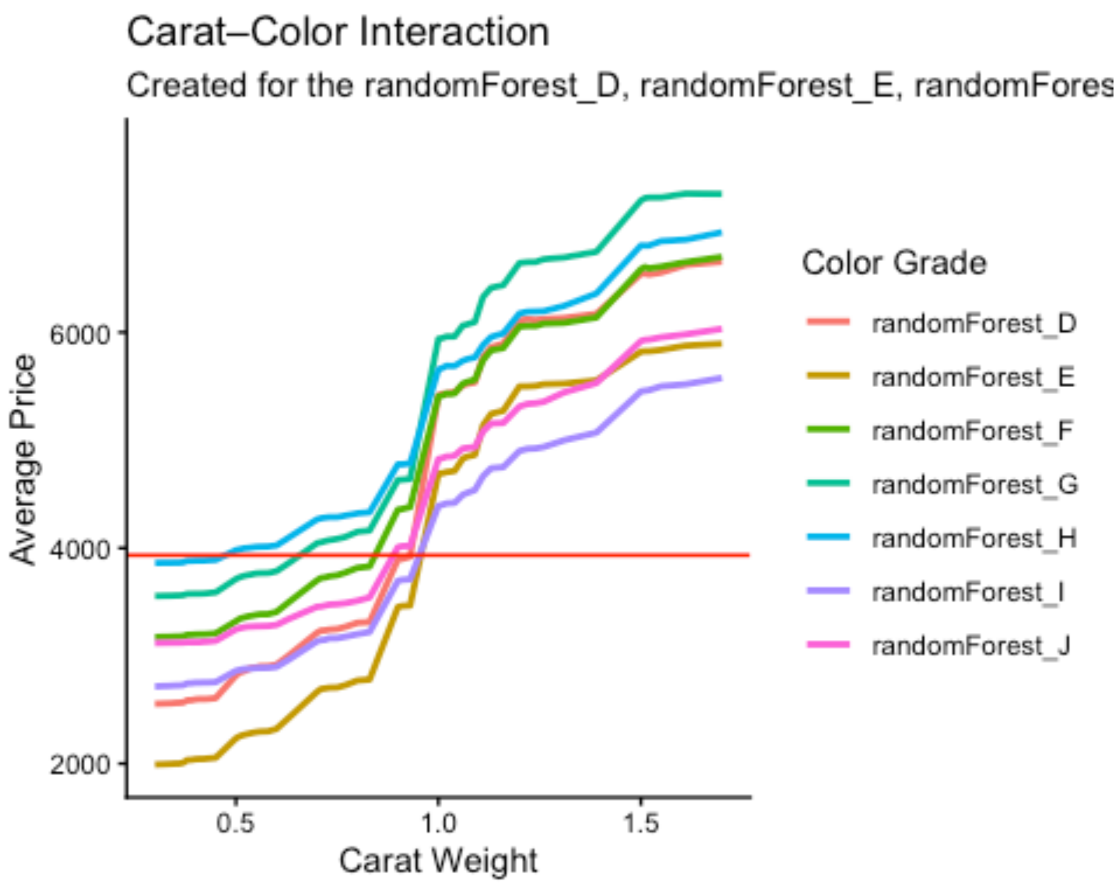



```r
# Examine interaction between num/cat or cat/cat variables (see variables in
numerics and cats created above)

set.seed(617904); interaction_profile <- model_profile(
  explainer = model_explainer,
  variables = c("carat"),
  groups = "color", type = "partial"
)

# Default
plot(interaction_profile)

## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## ℹ Please use tidy evaluation idioms with `aes()`.
## ℹ See also `vignette("ggplot2-in-packages")` for more information.
## ℹ The deprecated feature was likely used in the ingredients package.
##   Please report the issue at
##   <https://github.com/ModelOriented/ingredients/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
# Tweaked interaction plot
plot(interaction_profile) +
  labs(
    title = "Carat-Color Interaction",
    x = "Carat Weight",
    y = "Average Price"
  ) +
  lims(x = as.numeric(quantile(DATA$carat, c(.05, .95)))) +
  geom_hline(yintercept = mean(TRAIN.TARGET), linetype = "solid", color = "re
d", linewidth = 0.6) +
  scale_color_discrete(
    labels = c(
      "D" = "Color D - Highest Grade",
      "E" = "Color E",
      "F" = "Color F",
      "G" = "Color G",
      "H" = "Color H",
      "I" = "Color I",
      "J" = "Color J - Lowest Grade"
    ),
    name = "Color Grade"
  ) +
```

```
  theme_classic() +
  theme(strip.text.x = element_blank())

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

## Warning: Removed 56 rows containing missing values or values outside the s
cale range
## (`geom_line()`).
```




```
# Breakdown plot showing how particular prediction is pieced together
# Choose two types of rows that tell a good story
predictions <- predict(FOREST) #Get predicted value of each row
which(predictions >= quantile(predictions, .99))  #Some of the rows with high
est predictions

## 27311 27491 25276 27448 27330 27446 25436 26732 26753 27742 27402 25839 25
733
##    10    24   245   377   463   518   592   692   757  1071  1116  1156  1
225
## 27404 27499 27492 24927 26965 27397 26741 26820 27008 24663 25804 26374 26
833
##  1279  1366  1475  1621  1696  1710  1713  1721  1737  1751  1782  1812  1
856
```

```
## 26456 26887 25125 27079 27646 26926 27266 27649 27420 24189 25941 24476 27
739
##  2116  2189  2201  2253  2282  2343  2360  2640  2870  2891  2926  2959  3
164
## 27150 27219 27171 27629 27256 26935 27599 27066 26824 25763 27241 26268 27
198
##  3221  3533  3658  3721  4040  4097  4168  4327  4435  4501  4513  4605  4
614
## 26829 27243 27438 27525 27260 26446 27172 27332 25893 27632 27264 27750 27
355
##  4674  4689  4788  4975  5221  5253  5413  5523  5775  5950  6036  6197  6
365
## 25974 26260 27540 26933 27302 25533 26744 27522 26298 24674 27536 27400 27
556
##  6402  6464  6656  6819  6866  6881  6897  7071  7273  7299  8158  8240  8
249
## 26130 26566 27153 24166 26645 27278 25996 27455 27613 27480 27285 23775 26
914
##  8359  8516  8585  8629  8652  8767  8798  8843  8912  9107  9391  9425  9
430
## 26909 26883 27546 24946 27123 27098 27335 26612 27024 26326 26995 26855 27
014
##  9525  9602  9656  9665  9836  9877  9978 10014 10046 10114 10175 10225 10
242
## 27393 25017 26959 27728
## 10359 10392 10445 10758
```

```
which(predictions >= quantile(predictions, .49) & predictions <= quantile(pre
dictions, .51))  #Some of the rows with typical value of y
```

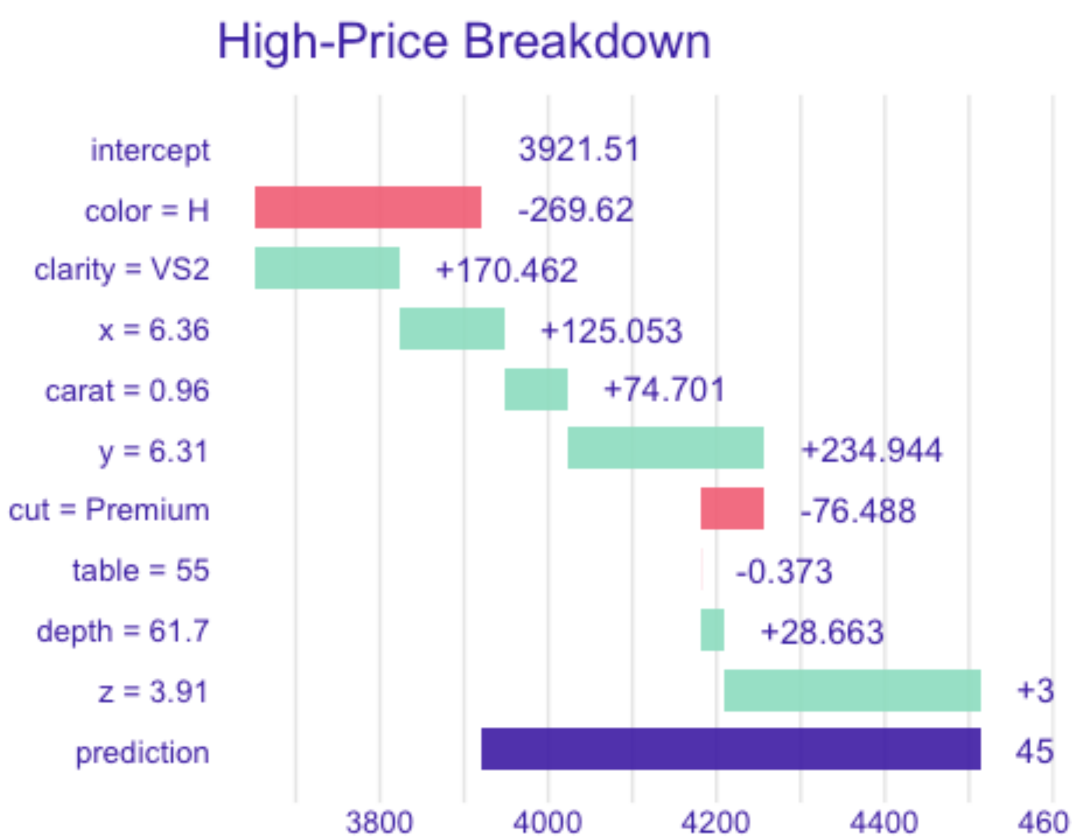
```
## 48944 51539 50552 52524 48798 49807 52735 52203 50857 53725 52088 53705 51
779
##     2    35    46    54   106   264   302   370   375   386   409   460
487
## 52179 49394 50749 51190 52398 52338 51018 49000 51449 51715 50579 52236 49
062
##   572   605   621   642   860   925   933   942  1007  1062  1114  1155  1
158
## 51295 53155 52694 48629   181 50055 51195 51540 51287 51047 51882 50170 52
018
##  1160  1179  1186  1237  1257  1308  1316  1324  1400  1420  1597  1764  1
841
## 49940 51594 50149 52794   292 53069 51486 50742 52821 50456 50806 52349 52
374
##  1998  2027  2064  2080  2229  2235  2344  2504  2522  2584  2588  2604  2
739
## 53160 53577 50588 49589 50039 48607 49720 52673 50388 49196 50719 49917 51
216
##  2748  2801  2812  2944  2946  2977  2984  3025  3309  3374  3398  3758  3
823
```

```
## 51597 50903 50403 51520 49576 49986 53154 51584 52430 49933  4693 53479 50
020
##  3834  3836  3837  3855  3887  3889  3901  3910  3923  3933  3936  4032  4
089
## 48411 52653 51073   151 50923 51952 52057 49326 51676 50870 50878 51749 52
848
##  4175  4188  4216  4285  4309  4369  4438  4440  4448  4462  4537  4538  4
654
## 50390 49499 51953 49402 50593 50690 52494 50867 50698 49368 52161 50444 48
920
##  4739  4774  4879  4886  4909  4982  5022  5042  5057  5085  5181  5227  5
251
## 53450 50772 51473 50431 52227 50941 50752 51864 51071 49790   208 49455 51
571
##  5313  5346  5420  5510  5524  5557  5652  5673  5747  5812  5820  5899  5
958
## 53191 49106 50881 48788 53077 47858 51469 52068 53255 50227 50751 49944 51
487
##  5981  6013  6025  6065  6133  6169  6198  6249  6425  6437  6454  6502  6
537
## 51116 51271 49796   553 52042 52993 50182 50217 53728 51213 49783 50555 50
333
##  6560  6619  6639  6919  6928  7044  7050  7054  7057  7119  7151  7190  7
262
## 48992 49922 50069 50154 50521 50065 52050 51157 48094 51617 47805 51616 53
706
##  7305  7370  7466  7468  7499  7510  7547  7553  7592  7728  7764  7826  7
847
## 50339 52498 51230 49721 51243 49875 50700 51096 51291 51686 52271 52266 51
375
##  7886  7945  7947  7955  8020  8261  8384  8532  8590  8671  8717  8849  8
871
## 53447 52326 51006 51665 52149 53559 51144 50930 52054 52693 52516 52138 47
782
##  8915  8940  8982  9031  9046  9085  9135  9161  9307  9314  9389  9393  9
419
## 51948 53315 52659 50219 51422 51388 51376 51532 48045 51234 48718 53932 52
945
##  9424  9427  9462  9467  9542  9564  9664  9687  9751  9752  9778  9945  9
966
## 50344 52798 52509 52268 53138 49985 48376 51386 51173 52504 52985 53681 50
147
##  9973 10139 10214 10240 10252 10256 10281 10328 10357 10361 10417 10433 10
505
## 45829 50520 53333 50422 49984 50386 49757 51787
## 10543 10596 10598 10675 10704 10705 10733 10781

which(predictions <= quantile(predictions, .01))  #Some of the rows with lowe
st predictions
```

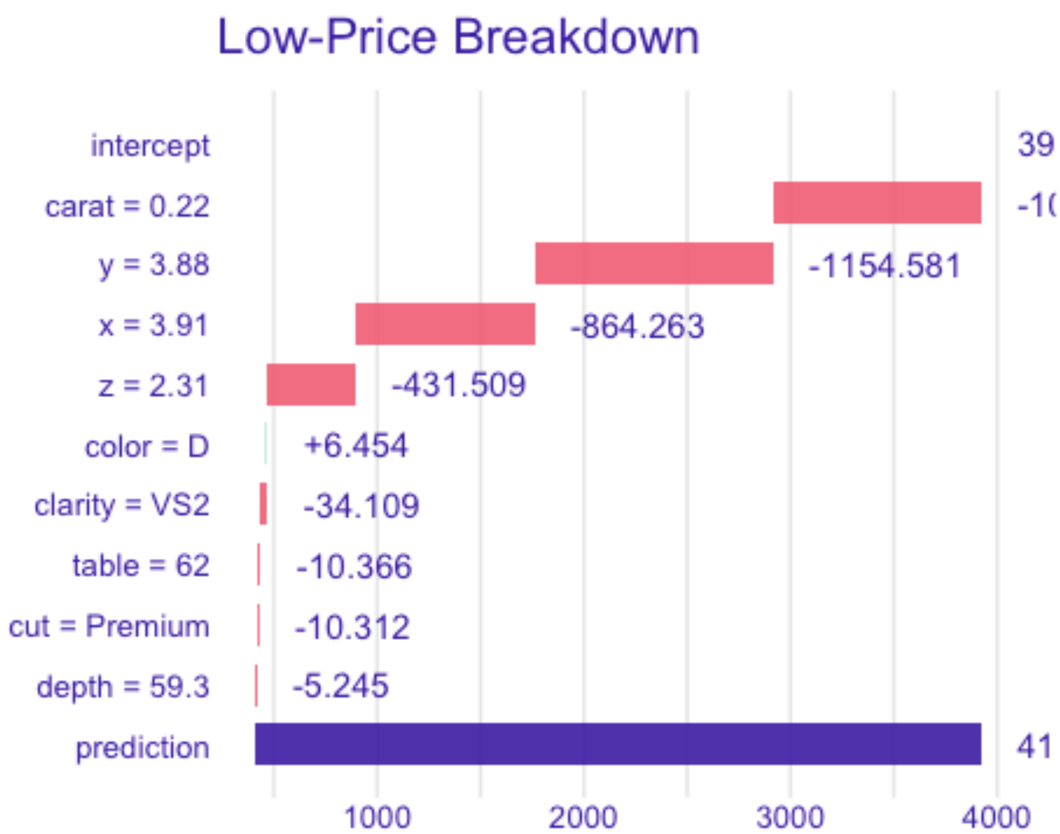
```
## 47283 37603 38254 38309 38263 16690 35639 30626 27050 34933 28650 33947 13
354
##   160   307   363   492   560   583   662   784  1085  1181  1421  1458  1
730
## 34262 31980 16695 37277 28631 29637 41601  6697 13355 31592 48305 30627
55
##  1840  1890  2203  2234  2340  2506  2585  2654  2782  2861  2957  3199  3
216
## 13365 28625 43927 47281 40607 32958  6721 13353 28277 13364 50637 41593 37
596
##  3252  3282  3362  3538  3649  3757  3783  3800  3893  3987  4015  4164  4
246
## 38269 31977 35969    60 13372 13047 43994 16691 29969    59 32282 28302 38
278
##  4304  4306  4382  4707  5011  5075  5168  5182  5348  5518  5552  5627  5
800
## 29307 34934 16703 43898 26691 27932 38252 29281 37274 31603 28972 13361 43
986
##  5845  5886  5966  6070  6098  6163  6215  6223  6363  6684  6728  6758  6
858
## 39946 47307 49961 42943 28980 26690 39608 20039 47308 48313 38275 37271 29
965
##  7169  7196  7203  7225  7277  7290  7714  7806  7919  7938  7989  8170  8
287
## 23364 28630 13370 28308 28954     9    51 10046 13360 43899  6708 34936 16
682
##  8304  8469  8480  8540  8586  8603  8768  8998  9008  9041  9049  9193  9
283
##  3378 47309 28309 28305  6704    22 32612 10045    36    37 23368 38257  3
384
##  9298  9450  9480  9491  9809  9825 10027 10051 10103 10110 10151 10152 10
221
## 34940 50632 29626 28634
## 10452 10464 10531 10784
```

```r
# Choosing a row with a high predicted price
# High-price diamond (row 9107)
bd_high <- predict_parts(
  explainer       = model_explainer,
  new_observation = DATA[9107, ],
  type            = "break_down"
)

plot(bd_high) +
  labs(
    title = "High-Price Breakdown"
  ) +
  theme(strip.text.x = element_blank())
```

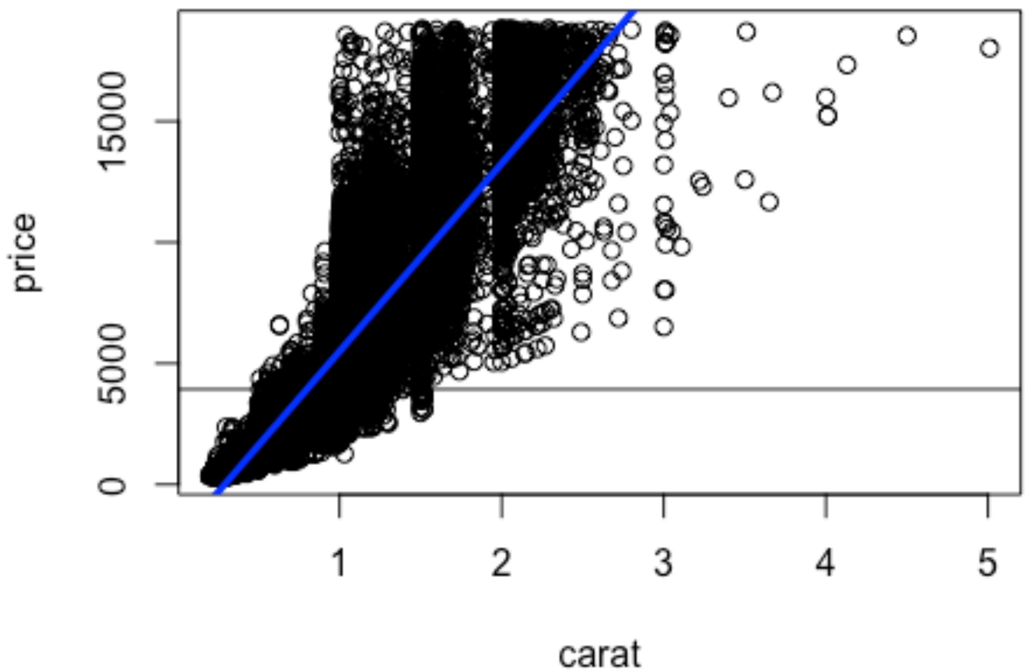
```
## `height` was translated to `width`.
```




```r
# Choosing a row with a lower predicted price
# Low-price diamond (row 55)
bd_low <- predict_parts(
  explainer       = model_explainer,
  new_observation = DATA[55, ],
  type            = "break_down"
)

plot(bd_low) +
  labs(
    title = "Low-Price Breakdown"
  ) +
  theme(strip.text.x = element_blank())

## `height` was translated to `width`.
```

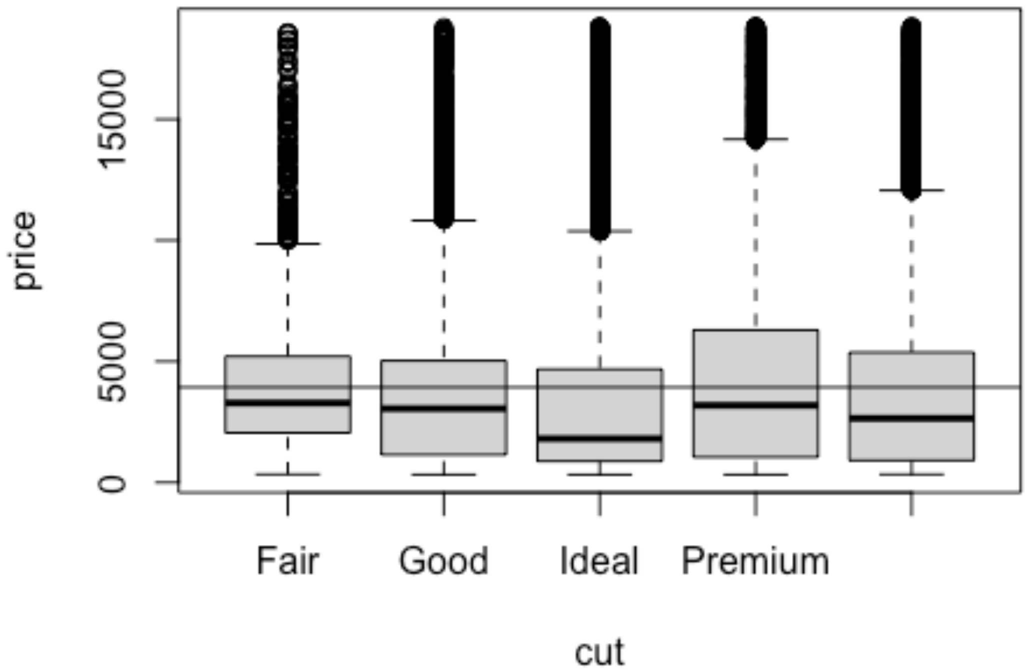



```
# Some of the relationships are nonlinear, so the "slope" doesn't a great job describing
# the relationship, and the driver scores may be off.  Can discretize!
examine_driver_Ynumeric(price ~ carat, data = DIAMONDS)
```

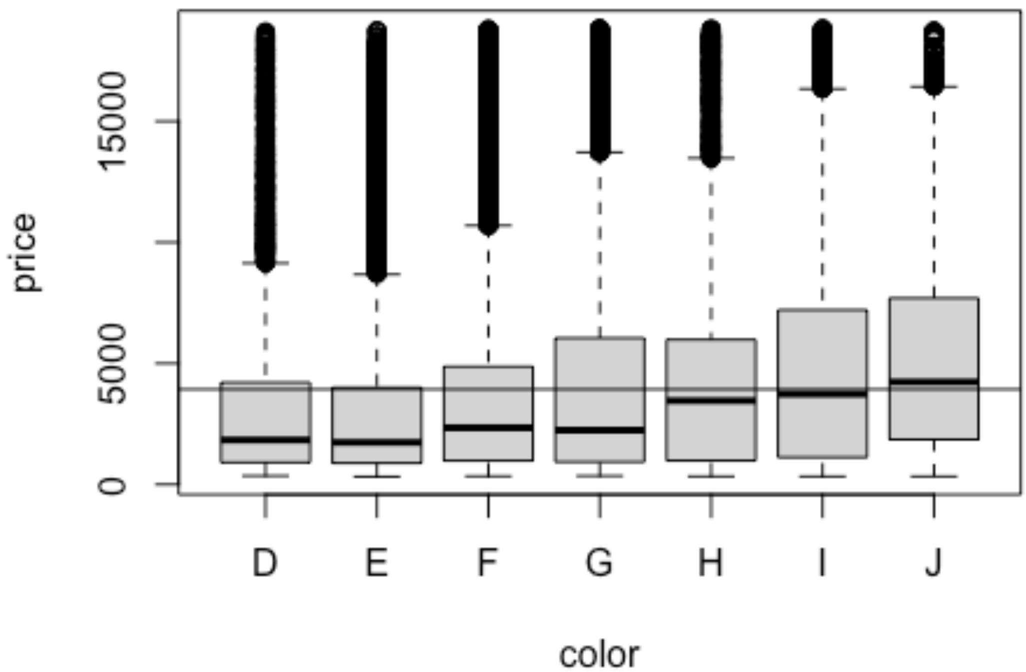



```
##
## Call:
## lm(formula = FORM, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18585.3   -804.8    -18.9    537.4  12731.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2256.36      13.06  -172.8   <2e-16 ***
## carat        7756.43      14.07   551.4   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1549 on 53938 degrees of freedom
## Multiple R-squared:  0.8493, Adjusted R-squared:  0.8493
## F-statistic: 3.041e+05 on 1 and 53938 DF,  p-value: < 2.2e-16
##
##
## Driver Score: 0.849
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
```

```
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.

examine_driver_Ynumeric(price ~ cut, data = DIAMONDS)
```

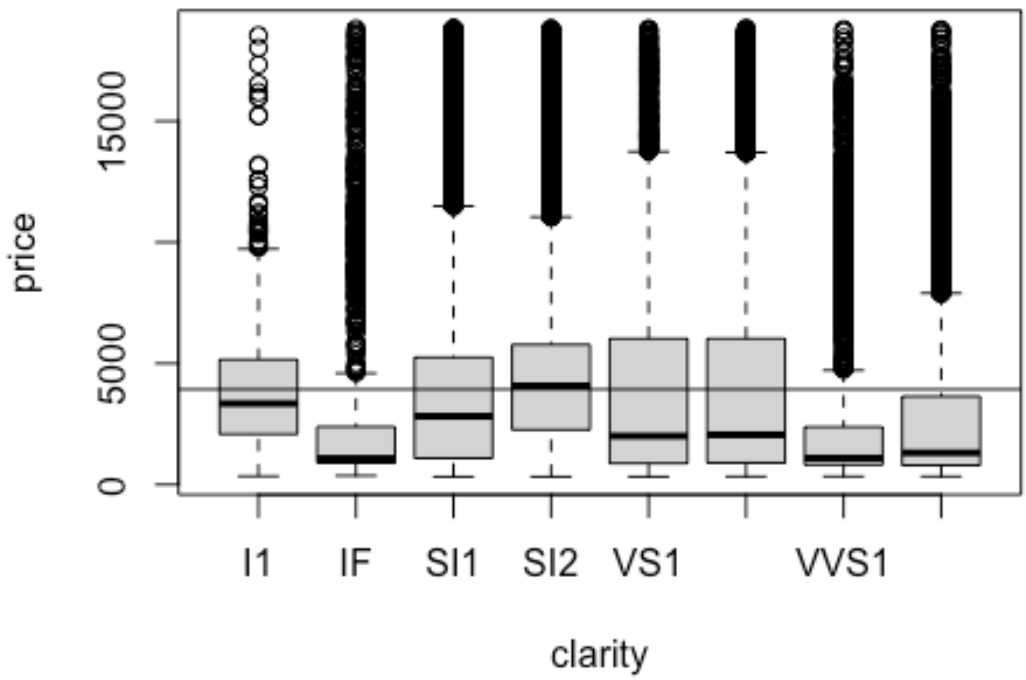



```
##          cut Avgprice letters     n
## 1   Premium 4584.258       a 13791
## 2      Fair 4358.758       a  1610
## 3 Very Good 3981.760       b 12082
## 4      Good 3928.864       b  4906
## 5     Ideal 3457.542       c 21551
##
## Driver Score: 0.013
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.

examine_driver_Ynumeric(price ~ color, data = DIAMONDS)
```

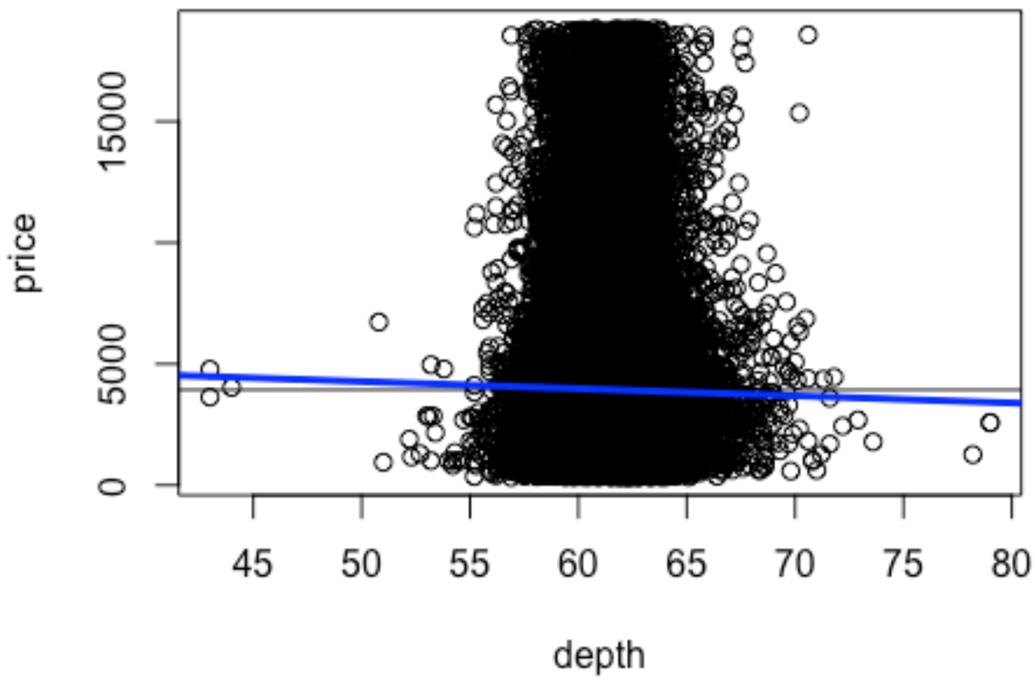



```
##   color Avgprice letters     n
## 1     J 5323.818       a  2808
## 2     I 5091.875       a  5422
## 3     H 4486.669       b  8304
## 4     G 3999.136       c 11292
## 5     F 3724.886       d  9542
## 6     D 3169.954       e  6775
## 7     E 3076.752       e  9797
##
## Driver Score: 0.031
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

```
examine_driver_Ynumeric(price ~ clarity, data = DIAMONDS)
```



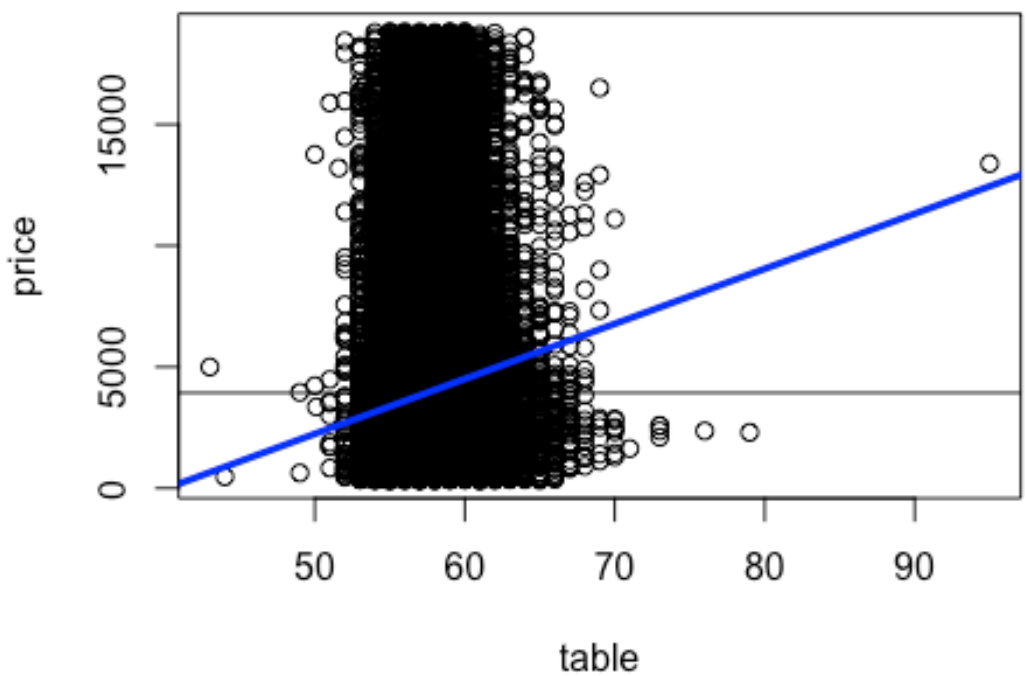

```
##   clarity Avgprice letters     n
## 1     SI2 5063.029       a  9194
## 2     SI1 3996.001       b 13065
## 3     VS2 3924.989       b 12258
## 4      I1 3924.169       b   741
## 5     VS1 3839.455       b  8171
## 6    VVS2 3283.737       c  5066
## 7      IF 2864.839       d  1790
## 8    VVS1 2523.115       d  3655
##
## Driver Score: 0.027
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.

examine_driver_Ynumeric(price ~ depth, data = DIAMONDS)
```




```
##
## Call:
## lm(formula = FORM, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -3766  -2986  -1521   1396  14937
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5763.67     740.56   7.783 7.21e-15 ***
## depth         -29.65      11.99  -2.473   0.0134 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3989 on 53938 degrees of freedom
## Multiple R-squared:  0.0001134,  Adjusted R-squared:  9.483e-05
## F-statistic: 6.115 on 1 and 53938 DF,  p-value: 0.0134
##
##
## Driver Score: 0
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
```

```
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

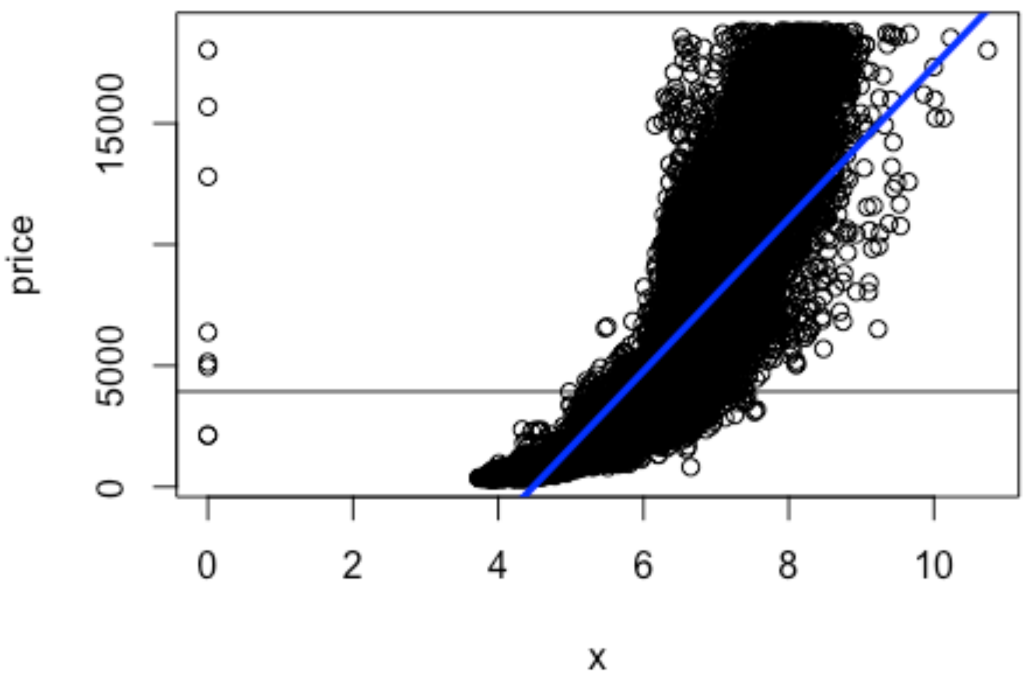
```r
examine_driver_Ynumeric(price ~ table, data = DIAMONDS)
```




```
##
## Call:
## lm(formula = FORM, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -6522  -2751  -1490   1368  15746
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9109.047    438.450  -20.78   <2e-16 ***
## table         226.984      7.625   29.77   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3957 on 53938 degrees of freedom
## Multiple R-squared:  0.01616,    Adjusted R-squared:  0.01614
## F-statistic: 886.1 on 1 and 53938 DF,  p-value: < 2.2e-16
```

```
##
##
## Driver Score: 0.016
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

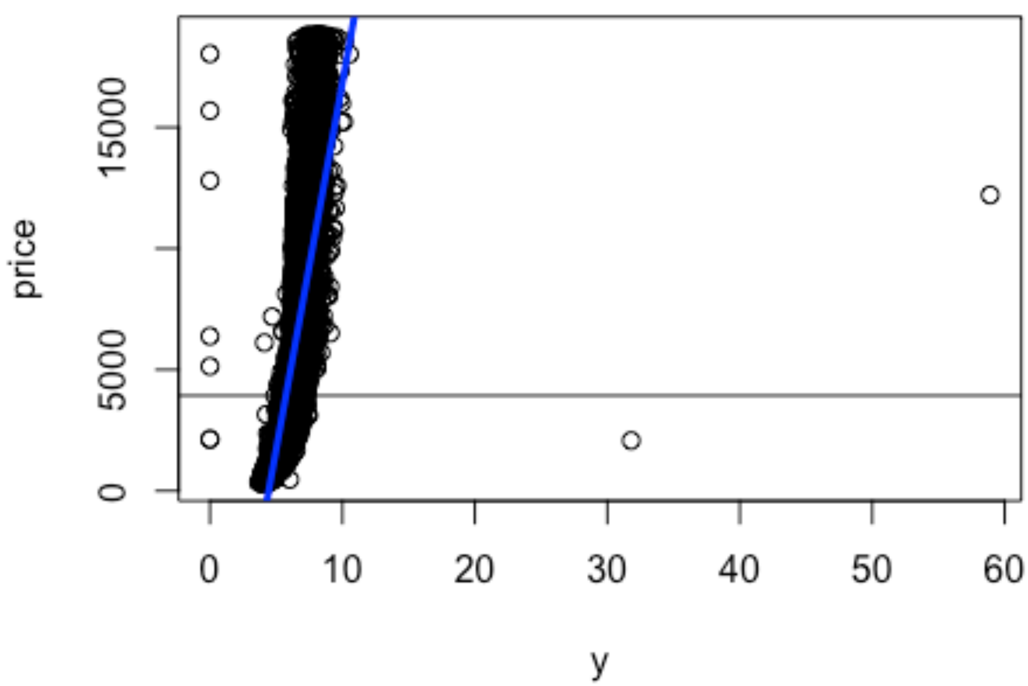
```r
examine_driver_Ynumeric(price ~ x, data = DIAMONDS)
```




```
##
## Call:
## lm(formula = FORM, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -8426  -1264   -185    973  32128
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept) -14094.056     41.732  -337.7   <2e-16 ***
## x             3145.413      7.146   440.2   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1862 on 53938 degrees of freedom
## Multiple R-squared:  0.7822, Adjusted R-squared:  0.7822
## F-statistic: 1.937e+05 on 1 and 53938 DF,  p-value: < 2.2e-16
##
##
## Driver Score: 0.782
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

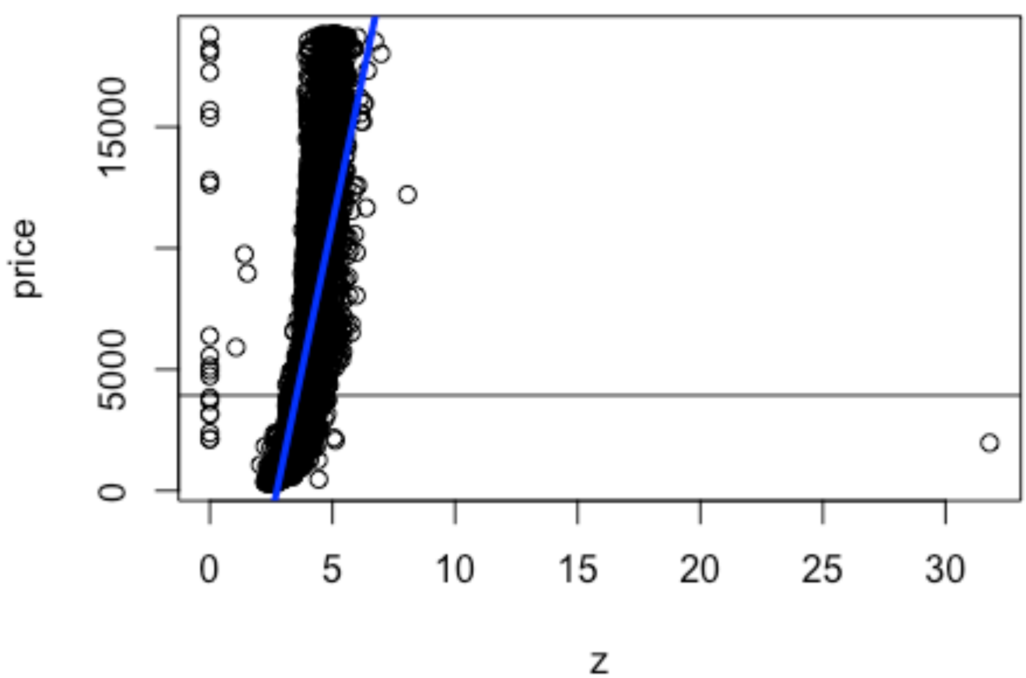
```r
examine_driver_Ynumeric(price ~ y, data = DIAMONDS)
```




```
##
## Call:
## lm(formula = FORM, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -152436   -1229    -241     838   31436
##
```

```
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13402.027     44.062  -304.2   <2e-16 ***
## y             3022.887      7.536   401.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1999 on 53938 degrees of freedom
## Multiple R-squared:  0.749,  Adjusted R-squared:  0.7489
## F-statistic: 1.609e+05 on 1 and 53938 DF,  p-value: < 2.2e-16
##
##
## Driver Score: 0.749
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

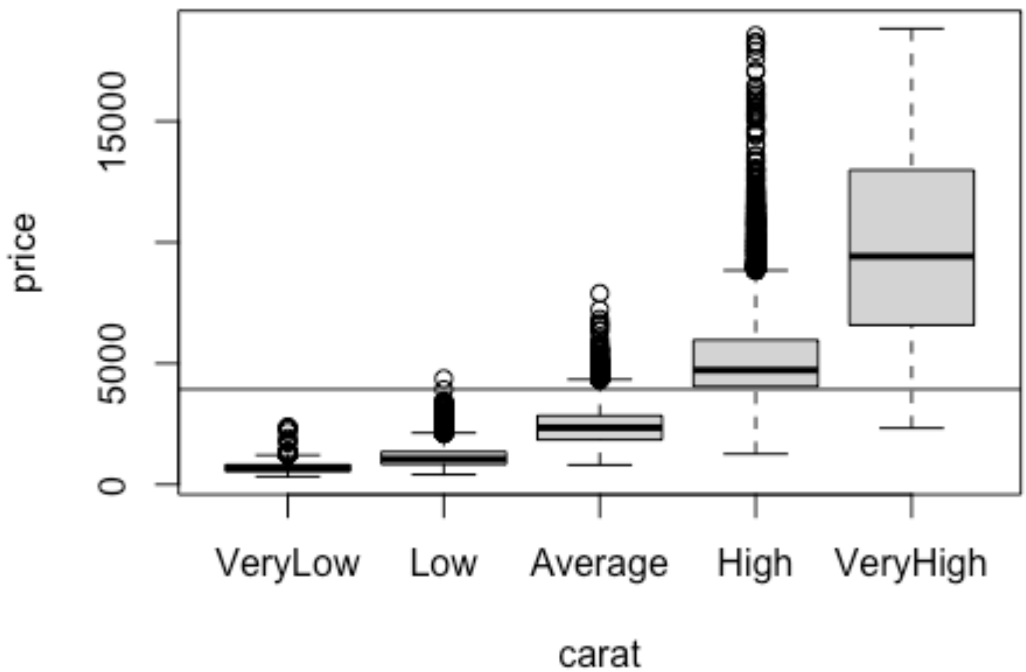
```r
examine_driver_Ynumeric(price ~ z, data = DIAMONDS)
```



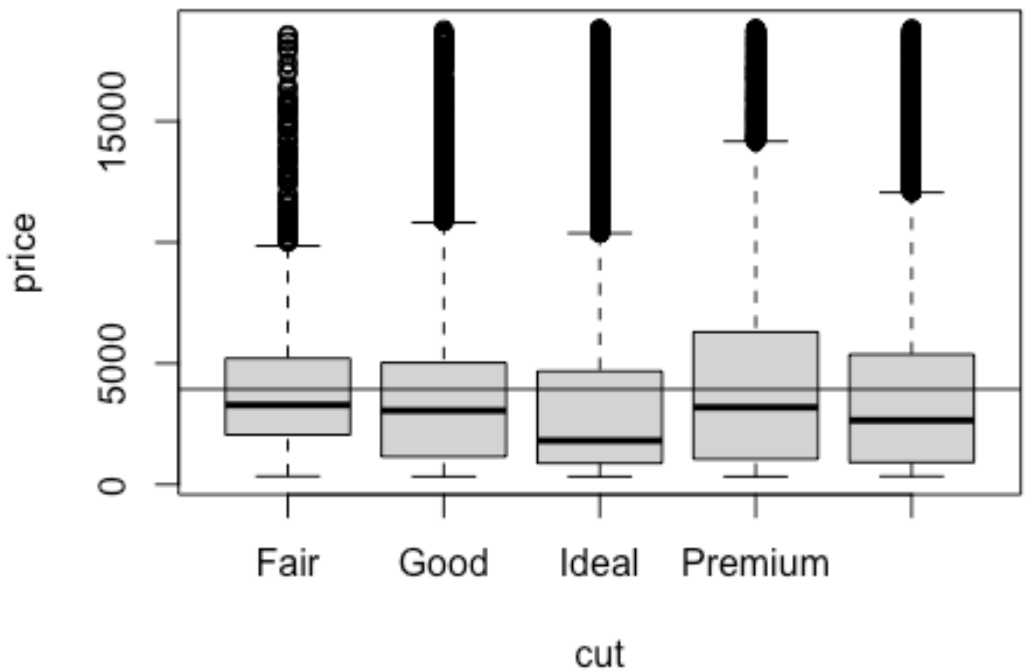

```
##
## Call:
## lm(formula = FORM, data = data)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -139561   -1235    -240     825   32085
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13296.57      44.64  -297.9   <2e-16 ***
## z             4868.79      12.37   393.6   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2027 on 53938 degrees of freedom
## Multiple R-squared:  0.7418, Adjusted R-squared:  0.7417
## F-statistic: 1.549e+05 on 1 and 53938 DF,  p-value: < 2.2e-16
##
##
## Driver Score: 0.742
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

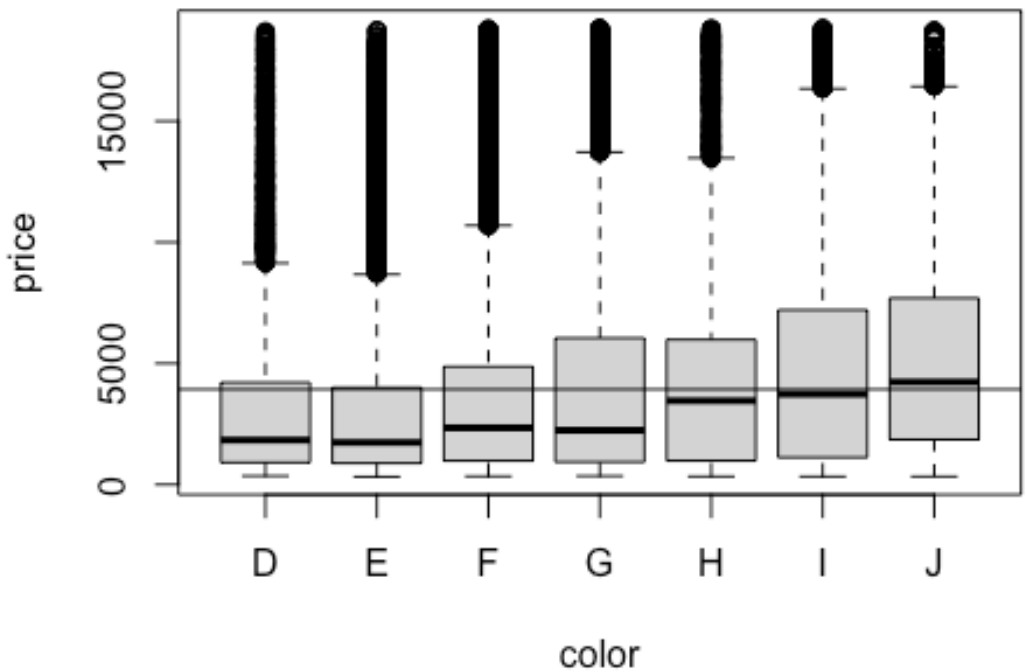
```r
# Discretize numeric drivers
# This will discretize ALL numeric columns, include price (which we don't wan
t)
DIAMONDS_DISC <- discretizeDF(
  DIAMONDS,
  default = list(
    method = "frequency",
    breaks = 5,
    include.lowest = TRUE,
    labels = c("VeryLow", "Low", "Average", "High", "VeryHigh")
  )
)
# Restore price column to its former glory
DIAMONDS_DISC$price <- DIAMONDS$price

examine_driver_Ynumeric(price ~ carat,   data = DIAMONDS_DISC)
```



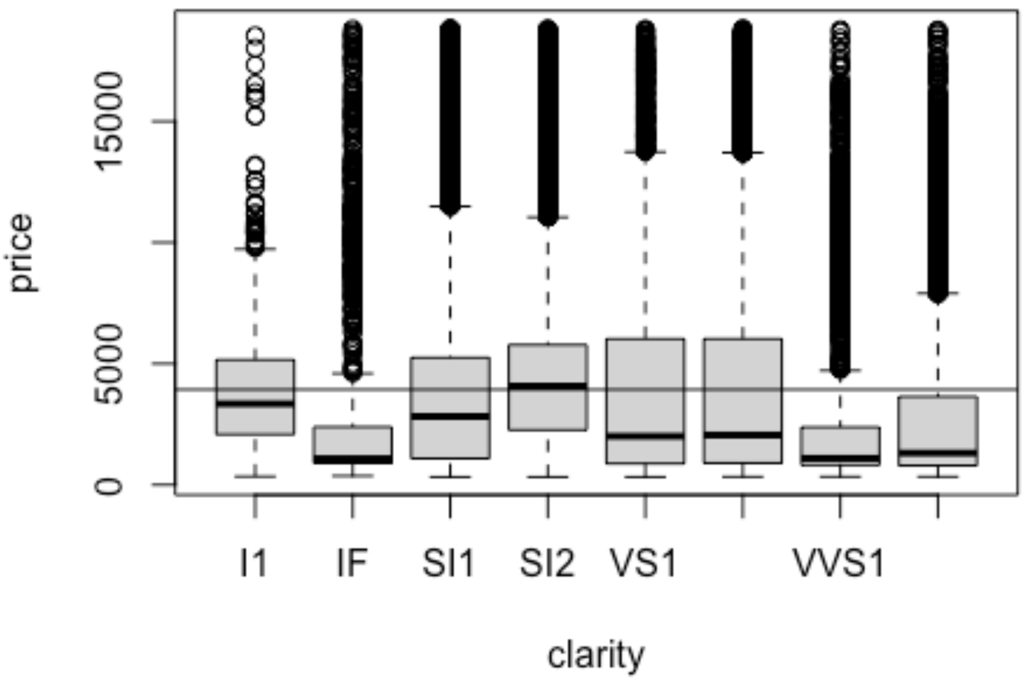

```
##       carat   Avgprice letters     n
## 1 VeryHigh 10021.2120       a 10933
## 2     High  5193.1124       b 10890
## 3  Average  2396.5353       c 11241
## 4      Low  1137.7278       d 10485
## 5  VeryLow   688.2671       e 10391
##
## Driver Score: 0.746
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.

examine_driver_Ynumeric(price ~ cut,     data = DIAMONDS_DISC)
```



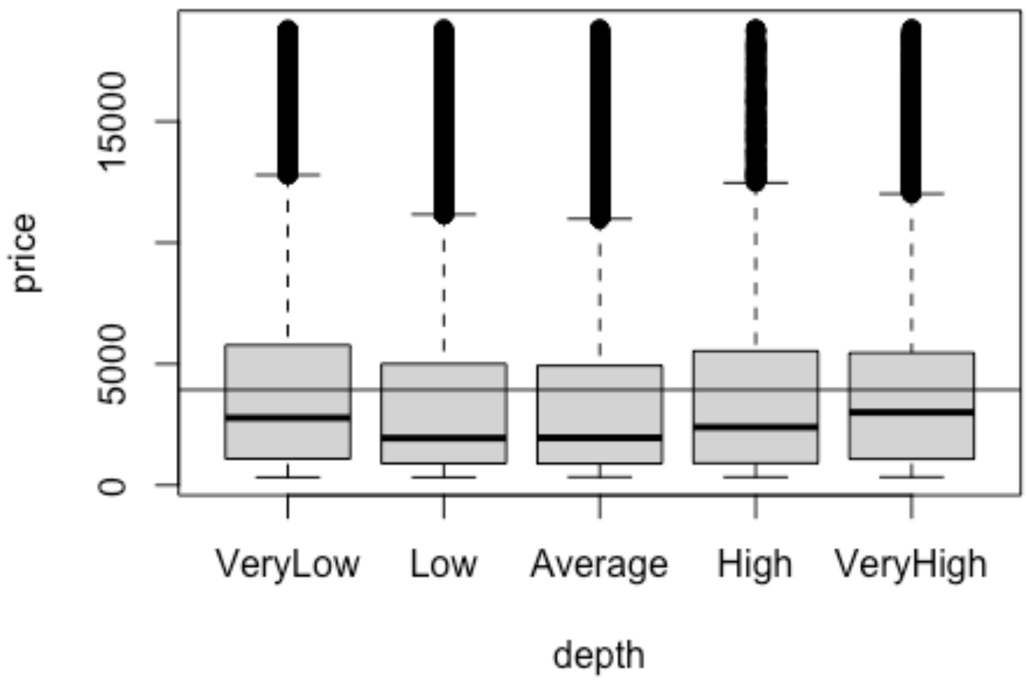

```
##          cut Avgprice letters     n
## 1   Premium 4584.258       a 13791
## 2      Fair 4358.758       a  1610
## 3 Very Good 3981.760       b 12082
## 4      Good 3928.864       b  4906
## 5     Ideal 3457.542       c 21551
##
## Driver Score: 0.013
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.

examine_driver_Ynumeric(price ~ color,   data = DIAMONDS_DISC)
```




```
##   color Avgprice letters     n
## 1     J 5323.818       a  2808
## 2     I 5091.875       a  5422
## 3     H 4486.669       b  8304
## 4     G 3999.136       c 11292
## 5     F 3724.886       d  9542
## 6     D 3169.954       e  6775
## 7     E 3076.752       e  9797
##
## Driver Score: 0.031
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.

examine_driver_Ynumeric(price ~ clarity, data = DIAMONDS_DISC)
```

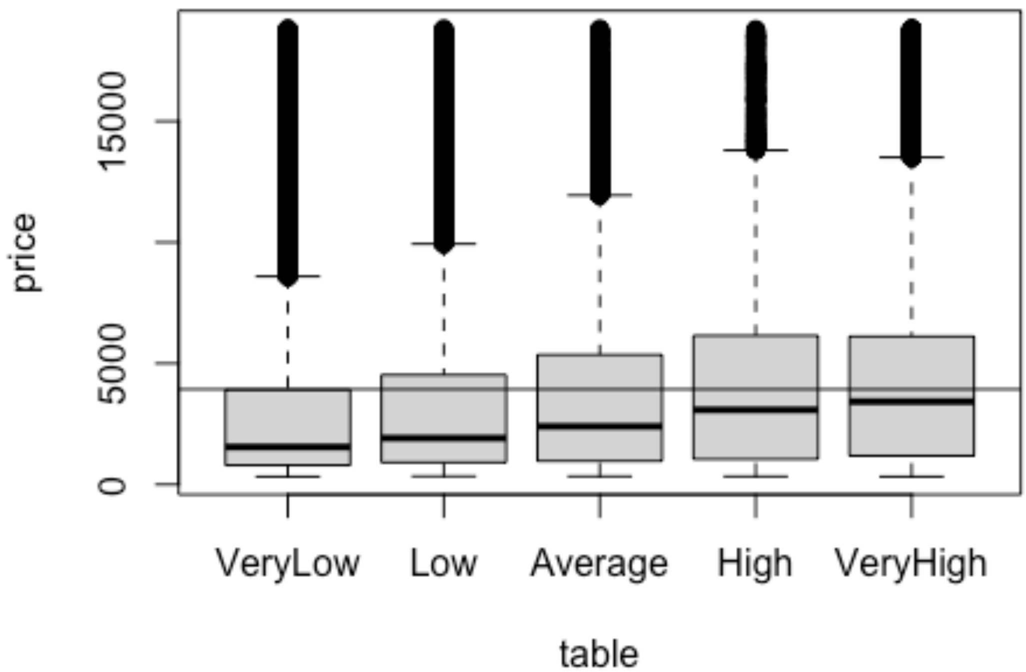



```
##   clarity Avgprice letters     n
## 1     SI2 5063.029       a  9194
## 2     SI1 3996.001       b 13065
## 3     VS2 3924.989       b 12258
## 4      I1 3924.169       b   741
## 5     VS1 3839.455       b  8171
## 6    VVS2 3283.737       c  5066
## 7      IF 2864.839       d  1790
## 8    VVS1 2523.115       d  3655
##
## Driver Score: 0.027
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.

examine_driver_Ynumeric(price ~ depth,   data = DIAMONDS_DISC)
```



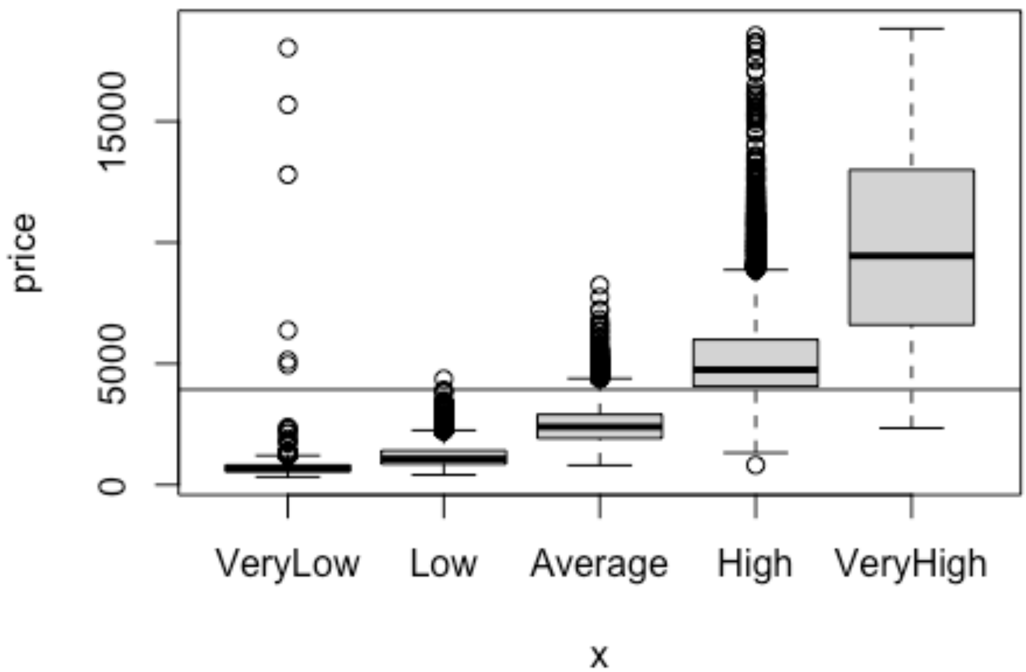

```
##      depth Avgprice letters     n
## 1  VeryLow 4330.643       a 10373
## 2 VeryHigh 4117.723       b 11579
## 3     High 3965.851       c 10851
## 4      Low 3656.408       d 10798
## 5  Average 3580.521       d 10339
##
## Driver Score: 0.005
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

```
examine_driver_Ynumeric(price ~ table,   data = DIAMONDS_DISC)
```

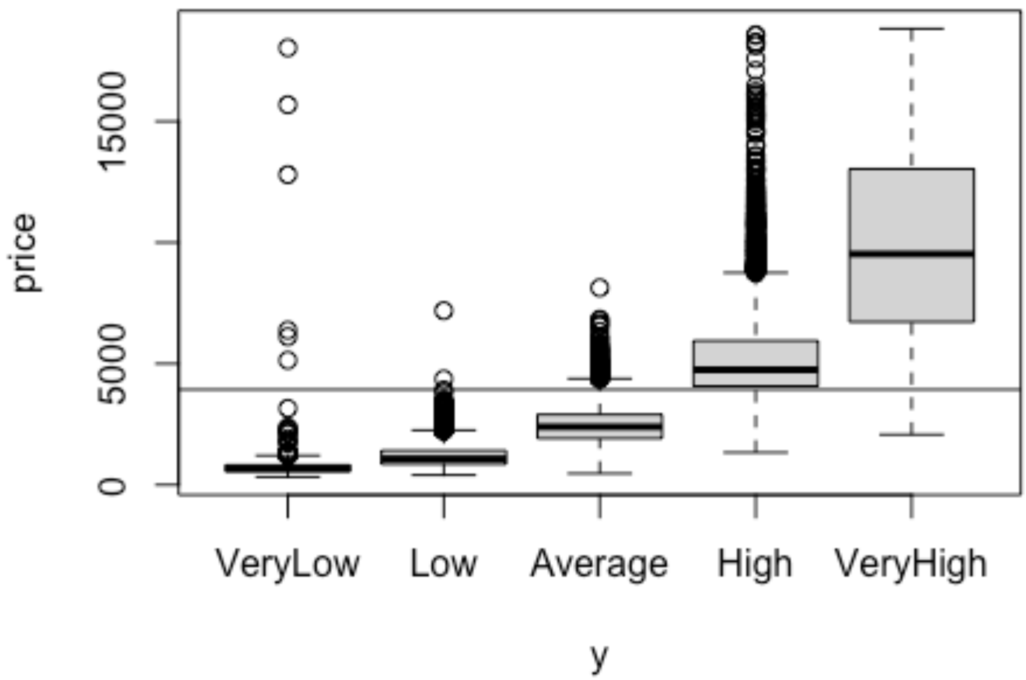



```
##      table Avgprice letters     n
## 1 VeryHigh 4599.181       a 15710
## 2     High 4408.966       b  8418
## 3  Average 3886.932       c  9805
## 4      Low 3384.872       d 10000
## 5  VeryLow 3078.577       e 10007
##
## Driver Score: 0.022
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

```
examine_driver_Ynumeric(price ~ x,       data = DIAMONDS_DISC)
```

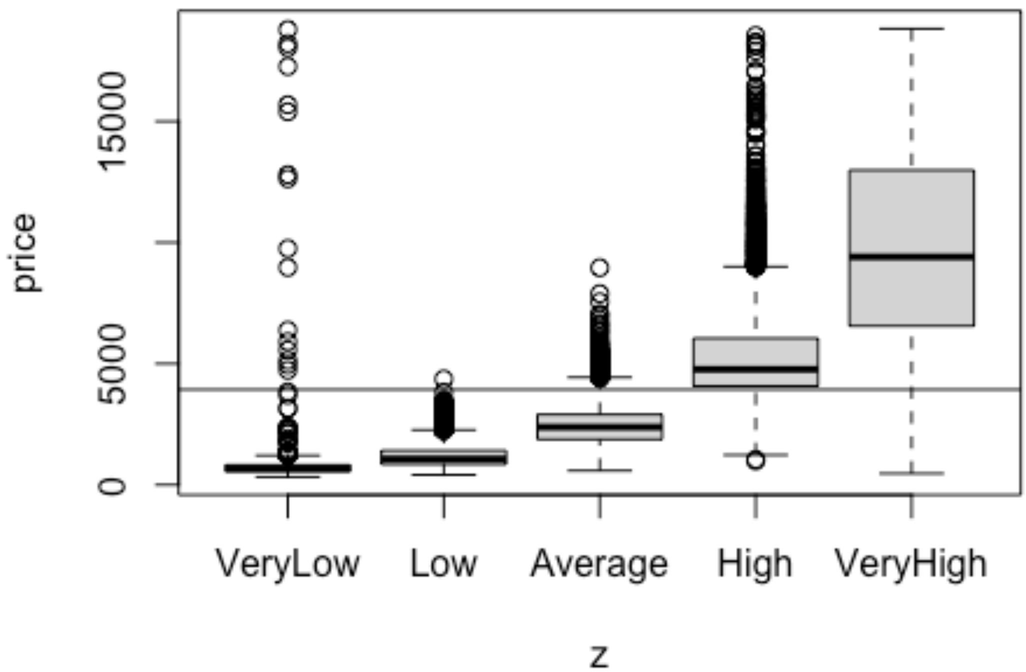



```
##          x  Avgprice letters     n
## 1 VeryHigh 10036.911       a 10877
## 2     High  5209.416       b 10804
## 3  Average  2463.028       c 10783
## 4      Low  1175.050       d 10842
## 5  VeryLow   694.234       e 10634
##
## Driver Score: 0.746
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.

examine_driver_Ynumeric(price ~ y,       data = DIAMONDS_DISC)
```




```
##           y   Avgprice letters     n
## 1 VeryHigh 10105.6541       a 10790
## 2     High  5197.4024       b 10835
## 3  Average  2459.6225       c 10825
## 4      Low  1174.5117       d 10811
## 5  VeryLow   698.4075       e 10679
##
## Driver Score: 0.752
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```

```
examine_driver_Ynumeric(price ~ z,       data = DIAMONDS_DISC)
```




```
##          z  Avgprice letters     n
## 1 VeryHigh 9990.8717       a 10901
## 2     High 5241.2129       b 10702
## 3  Average 2448.5024       c 11100
## 4      Low 1171.1724       d 10538
## 5  VeryLow  711.5687       e 10699
##
## Driver Score: 0.739
## Caution: misleading if driver is numerical and trend isn't linear.
## Scores range between 0 and 1.  Larger scores = stronger driver.
## Although context dependent, values above 0.02 or so are 'reasonably strong
' drivers.
```